

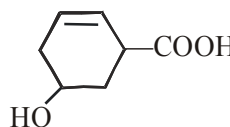
ORGANIC CHEMISTRY

S.No.	CONTENTS	Page
1.	Classification & Nomenclature System	1
2.	Isomerism	6
3.	General Organic Chemistry	12
4.	Hydrocarbon & Aromatic Hydrocarbon	20
5.	Halogen and its derivative	29
6.	Alcohol, Phenol and Ether	36
7.	Carbonyl Compounds and Carboxylic acid & its derivative	43
8.	Nitrogen Containing Compounds	50
9.	Biomolecules, Polymer and POC & Chemistry in everyday life	57

CLASSIFICATION & NOMENCLATURE SYSTEM

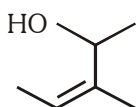
EXERCISE-I

1.  has IUPAC name :-



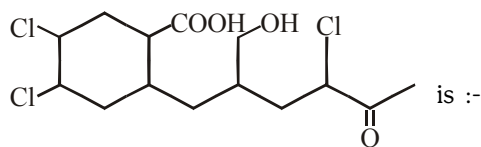
- (1) 3-hydroxy cyclohex-5-en carboxylic acid
- (2) 5-hydroxy cyclohex-2-en carboxylic acid
- (3) 5-hydroxy cyclohexanoic acid
- (4) 3-hydroxy cyclohex-5- enoic acid

- 2.** The IUPAC name of the compound  is:-



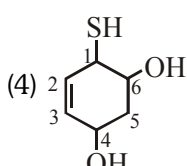
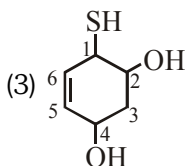
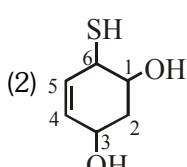
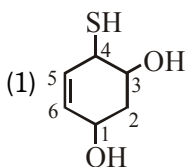
- (1) 3, 4-Dimethylbut-2-en-4-ol
- (2) 3-Methylpent-2-en-4-ol
- (3) 3-Methylpent-3-en-2-ol
- (4) 1, 2-Dimethylbut-2-en-1-ol

- 3.** The correct IUPAC name of

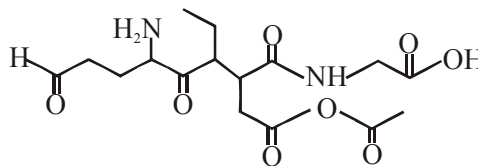


- (1) 3, 4 dichloro-6-[4-chloro-2-(hydroxymethyl)-5-oxohexyl] cyclohexane -1- carboxylic acid
- (2) 4,5-dichloro-2-[4-acetyl-4-chloro-2-(hydroxymethyl butyl)] cyclohexane -1- carboxylic acid.
- (3) 4-5-dichloro-2-[4-chloro-2-(hydroxymethyl)-5-oxohexyl] cyclohexane -1- carboxylic acid.
- (4) None of the above

4. Select the structure with correct numbering for IUPAC name of the compound -



5. The number of functional groups present in the following compound is :-



- (1) 5 (2) 7 (3) 6 (4) 8

- 6.** Alicyclic compounds are

- (1) Aromatic compounds
- (2) Aliphatic cyclic compounds
- (3) Heterocyclic compounds
- (4) None of the above

7. How many 1°, 2°, 3° C atoms does 1, 3, 5-Trimethylcyclohexane have?

- (1) 3, 6, 0 (2) 3, 4, 2
(3) 0, 3, 6 (4) 3, 3, 3

- 8.** The compound which has one isopropyl group is:

- (1) 2,2,3,3-Tetramethyl pentane
- (2) 2,2-Dimethyl pentane
- (3) 2,2,3-Trimethyl pentane
- (4) 2-Methyl pentane

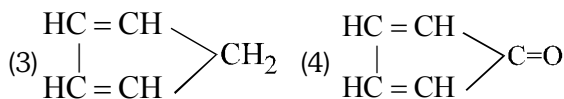
9. Which of the following is the first member of ester homologous series?

- (1) Ethyl ethanoate (2) Methyl ethanoate
(3) Methyl methanoate (4) Ethyl methanoate

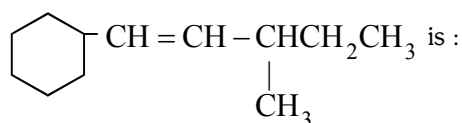
- 10.** The molecular formula of the first member of the family of alkenynes and its name is given by the set

- (1) C_3H_6 , Alkene
- (2) C_5H_6 , Pent-1-en-3-yne
- (3) C_6H_8 , Hex-1-en-5-yne
- (4) C_4H_4 , Butenyne

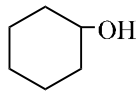
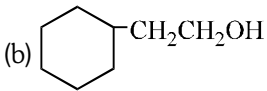
- 11.** Which of the following is a heterocyclic compound



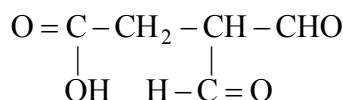
- 12.** The IUPAC name of



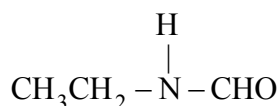
- (1) 1-Cyclohexyl-3-methylpent-1-ene
- (2) 3-Methyl-5-cyclohexylpent-1-ene
- (3) 1-Cyclohexyl-3-ethylbut-1-ene
- (4) 1-Cyclohexyl-3,4-dimethyl-but-1-ene

13. IUPAC name of $\text{CH}_2=\text{CH}-\text{CN}$ is:
 (1) Ethenenitrile (2) Vinyl cyanide
 (3) Cyono ethene (4) Prop-2-enenitrile
14. The IUPAC name of $\text{CH}_3\text{CH}_2-\text{N}(\text{CH}_3)-\text{CH}_2\text{CH}_3$ is:
 (1) N-Methyl-N-ethyl ethanamine
 (2) Diethyl methanamine
 (3) N-Ethyl-N-methyl ethanamine
 (4) Methyl diethyl ethanamine
15. (a)  and (b) 
 True statement for the above compounds is:
 (1) (a) is phenol while (b) is alcohol
 (2) Both (a) and (b) are primary alcohol
 (3) (a) is primary and (b) is secondary alcohol
 (4) (a) is secondary and (b) is primary alcohol
16. The IUPAC name of $(\text{C}_2\text{H}_5)_2\text{NCH}_2\text{CH}(\text{Cl})\text{COOH}$ is
 (1) 2-Chloro-4-N-ethylpentanoic acid
 (2) 2-Chloro-3-(N,N-diethyl amino)-propanoic acid
 (3) 2-Chloro-2-oxo diethylamine
 (4) 2-Chloro-2-carboxy-N-ethyl ethane
17. Which of the following compound is wrongly named?
 (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{Cl})\text{COOH}$;
 2-Chloro pentanoic acid
 (2) $\text{CH}_3\text{C}(\text{CH}_3)=\text{CHCOOH}$;
 2-Methyl hex-3-enoic acid
 (3) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCOCH}_3$;
 Hex-3-en-2-one
 (4) $\text{CH}_3-\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CHO}$;
 4-Methyl pentanal

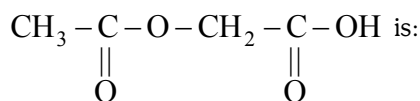
18. The correct IUPAC name of the following compound is:



- (1) 3,3-Diformylpropanoic acid
 (2) 3-Formyl-4-oxo-butanoic acid
 (3) 3,3-Dioxo propanoic acid
 (4) 3,3-Dicarbaldehyde propanoic acid
19. One among the following is the correct IUPAC name of the compound

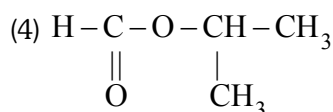
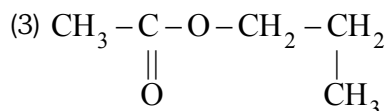
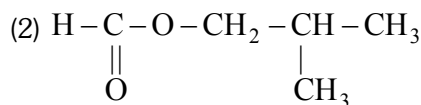
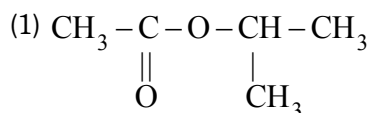


- (1) N-Formyl aminoethane
 (2) N-Ethyl formyl amine
 (3) N-Ethyl methanamide
 (4) Ethylamino methanal
20. The IUPAC name of $\text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{COOH}$ is:
 (1) Cinnamic acid
 (2) 1-Phenyl-2-carboxy ethane
 (3) 3-Phenyl prop-2-enoic acid
 (4) Dihydroxy-3-phenyl propionic acid
21. IUPAC name will be $\text{CH}_2(\text{CN})-\text{CH}(\text{CN})-\text{CH}_2(\text{CN})$
 (1) 1,2,3-Tricyano propane
 (2) Propane-1,2,3- trinitrile
 (3) 1,2,3-Cyano propane
 (4) Propane-1,2,3-tricarbonitrile
22. The IUPAC name of



- (1) 1-Acetoxy acetic acid
 (2) 2-Acetoxy ethanoic acid
 (3) 2-Ethanoyloxyacetic acid
 (4) 2-Ethanoyloxyethanoic acid
23. The IUPAC name of $\text{OHC}-\text{CH}_2-\text{CH}(\text{CH}_2-\text{CHO})-\text{CH}_2-\text{CHO}$ is:
 (1) 4,4-Di(formylmethyl) butanal
 (2) 2-(Formylmethyl) butane-1, 4-dicarbaldehyde
 (3) Hexane-3-acetal-1, 6-dial
 (4) 3-(Formylmethyl) hexane-1, 6-dial

24. Structural formula of isopropyl methanoate is:



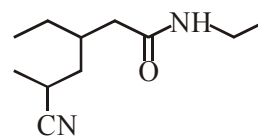
25. The correct order of priority of the functional groups used as suffix will be :-

- (1) $-\text{COOH} > -\text{CHO} > -\text{CN} > \text{C} = \text{O}$
 (2) $-\text{COOH} > -\text{CHO} > \text{C} = \text{O} > -\text{CN}$
 (3) $>\text{C} = \text{O} > -\text{CN} > -\text{COOH} > -\text{CHO}$
 (4) $-\text{COOH} > -\text{CN} > -\text{CHO} > \text{C} = \text{O}$

26. For which of the following compounds, the IUPAC name is wrong?

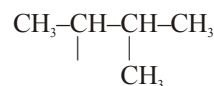
Compound	IUPAC Name
(1)	2-methyl cyclopentane carboxamide
(2)	Cyclohexanoyl chloride
(3)	2-methyl cyclobutane carbonitrile
(4)	Methyl-2 bromo cyclohexane carboxylate

27. What is the correct IUPAC name of the following compound?



- (1) 3-(2-cyanopropyl)-N-ethyl pentanamide
 (2) N-ethyl-3-(2-cyanopropyl) pentanamide
 (3) N,3-diethyl-4-cyano hexanamide
 (4) 5-cyano-N, 3-diethyl hexanamide

28. Which of the following correct IUPAC name of:-



- (1) 1,1,-Dimethylpropyl
 (2) 1,2-Dimethylpropyl
 (3) Active iso sec pentyl
 (4) 3-Methyl butyl

29. IUPAC name of $\text{CH}_2 = \text{CH}-\text{CH}_2-$ is :-

- (1) 1-Propene (2) 1-Propenyl
 (3) 2-Propenyl (4) Allyl

30. Number of σ and π bond in phenyl cyanide respectively :-

- (1) 7σ and 2π (2) 12σ and 5π
 (3) 8σ and 2π (4) 13σ and 5π

31. Which of the following structure contain aldehyde functional group :-

- (1) $\text{CH}_3-\text{NH}-\underset{\text{H}}{\text{C}}=\text{O}$ (2) $\text{Cl}-\underset{\text{H}}{\text{C}}=\text{O}$
 (3) $\text{H}_2\text{N}-\text{CH}_2-\underset{\text{H}}{\text{C}}=\text{O}$ (4) All of these

32. IUPAC name is :-

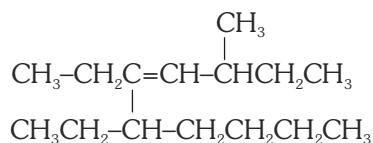
- (1) 2-Methyl pent-3-ene
 (2) 4-Methyl pent-2-ene
 (3) 3-Methyl pent-2-ene
 (4) 3-Methyl pent-3-ene

ANSWER KEY						Exercise-I				
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	2	3	2	4	4	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	4	3	4	2	2	2	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	4	4	4	2	4	2	3	4
Que.	31	32								
Ans.	3	2								

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

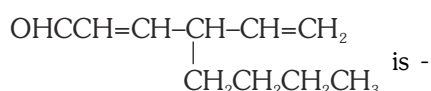
1. The correct IUPAC name of the compound is-



[IIT-90]

- (1) 5,6-diethyl-8-methyl dec-6-ene
(2) 5,6-diethyl-3-methyl dec-4-ene
(3) 6-butyl-5-ethyl-3-methyl oct-4-ene
(4) 2,4,5-triethyl-3-ene

2. The IUPAC name of - [IIT-91]

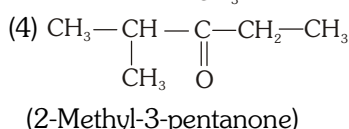
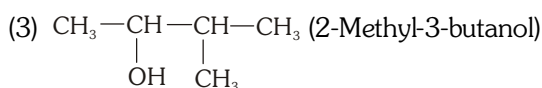
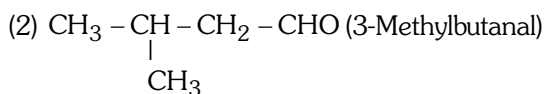


- (1) 5-Vinyloct-3-en-1-al
(2) 4-Butylhexa-2,5-dien-1-al
(3) 5-Vinyloct-5-en-8-al
(4) 3-Butylhexa-1,4-dien-8-al

3. IUPAC nomenclature of the given organic compound $(\text{CH}_3)_2\text{C}(\text{CH}_2\text{CH}_3)\text{CH}_2\text{CH}(\text{Cl})\text{CH}_3$ will be - [Bihar CEE 2001]

- (1) 5-Chloro-3,3-dimethyl hexane
(2) 4-Chloro-2-ethyl-2-methylpentane
(3) 2-Chloro-4-ethyl-4-methylpentane
(4) 2-Chloro-4,4-dimethylhexane

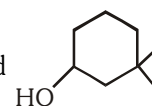
4. Which of the following compound has wrong IUPAC name? [AIEEE -2002]



5. The IUPAC name of $\text{CH}_3\text{COCH}(\text{CH}_3)_2$ is [AIEEE-2003]

- (1) isopropyl methyl ketone
(2) 2-methyl-3-butanone
(3) 4-methylisopropyl ketone
(4) 3-methyl-2-butanone

6. The IUPAC name of the compound is



[AIEEE-2004]

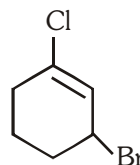
- (1) 1, 1-dimethyl-3-cyclohexanol
(2) 1, 1-dimethyl-3-hydroxy cyclohexane
(3) 3, 3-dimethyl-1-cyclohexanol
(4) 3, 3-dimethyl-1-hydroxy cyclohexane

7. Choose the correct molecular formula of the following molecules : [VITEEE-2005]

(i) Propanone, (ii) Propanal and (iii) n-propanol

- (1) CH_3COCH_3 , $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and $\text{CH}_3\text{CH}_2\text{CHO}$
(2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$, CH_3COCH_3 and $\text{CH}_3\text{CH}_2\text{CHO}$
(3) $\text{CH}_3\text{CH}_2\text{CHO}$, CH_3COCH_3 and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
(4) CH_3COCH_3 , $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

8. The IUPAC name of the compound is



[AIEEE-2006]

- (1) 6-bromo-2-chlorocyclohexene
(2) 3-bromo-1-chlorocyclohexene
(3) 1-bromo-3-chlorocyclohexene
(4) 2-bromo-6-chlorocyclohex-1-ene

9. IUPAC name of $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$ [IIT-2006]

- (1) Benzoylchloride
(2) Benzenecarbonylchloride
(3) Chlorophenyl ketone
(4) Phenylchloroketone

10. The correct decreasing order of priority for the functional groups of organic compounds in the IUPAC system of nomenclature is [AIEEE -2008]

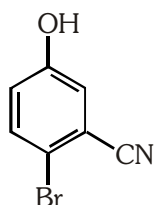
- (1) $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$, $-\text{CHO}$
(2) $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{CONH}_2$, $-\text{CHO}$
(3) $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$
(4) $-\text{CONH}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$

11. The IUPAC name of neopentane is :-

[AIEEE -2009]

- (1) 2-methylpropane
- (2) 2, 2-dimethylbutane
- (3) 2-methylbutane
- (4) 2, 2-dimethylpropane

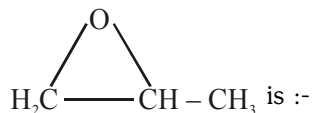
12. The IUPAC name of the following compound is :



[IIT-JEE-2009]

- (1) 4-Bromo-3-cyanophenol
- (2) 2-Bromo-5-hydroxybenzonitrile
- (3) 2-Cyano-4-hydroxybromobenzene
- (4) 6-Bromo-3-hydroxybenzonitrile

13. The IUPAC name of the compound



[AIEEE-2012 (Online)]

- (1) 1, 2-Epoxy propane
- (2) Propylene oxide
- (3) 1, 2-Oxo propane
- (4) 1, 2-Propoxide

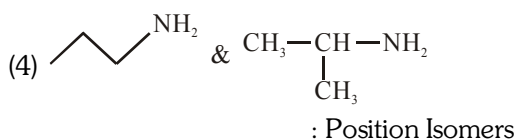
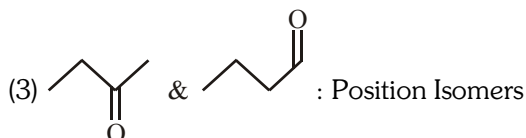
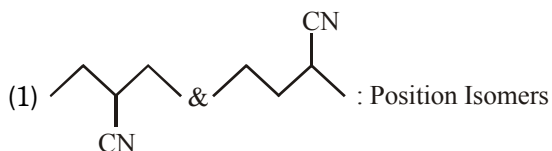
PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4	3	4	3	4	2	2	1
Que.	11	12	13							
Ans.	4	2	1							

ISOMERISM

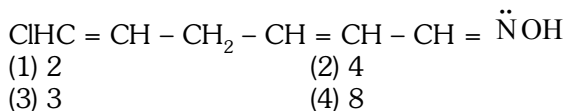
EXERCISE-I

1. Molecular weight of a hydrocarbon containing minimum number of C-atom to show optical isomerism:-
 (1) 100 (2) 80 (3) 68 (4) 70
2. Compounds which can show both optical as well as geometrical isomerism :
- (1)
- (2)
- (3)
- (4)
3. Which of the following will not show optical isomerism :-
- (1)
- (2)
- (3)
- (4)
4. Relation between given compound
-
- (1) Both are identical (2) Diastereomer
 (3) Enantiomer (4) Geometrical isomer
5. Which of the following structure is chiral ?
- (1)
- (2)
- (3)
- (4)
6. The No. of optically active stereoisomer of:-
 $\text{CH}_3\text{-CH=CH-CH(OH)-CH=CH-CH}_3$
 (1) 2 (2) 3
 (3) 4 (4) 6
7. Which of the following compounds are chiral?
- (1)
- (2)
- (3)
- (4) All of these
8. Which does not show geometrical isomerism?
- (1)
- (2)
- (3)
- (4)
9. The correct statement about the compounds I, II and III.
- (I)
- (II)
- (III)
- (1) I and II are identical
 (2) I and II are diastereomers
 (3) I and III are enantiomers
 (4) I and II and enantiomers
10. Which of the following compounds can exist in *meso* form ?
 (1) 2, 3-Pentanediol (2) 2, 3-Butanediol
 (3) 1, 2-Ethanediol (4) 1, 2-Propanediol

11. Which of the following is correct matching :-



12. The number of geometrical isomers for following structure is :-



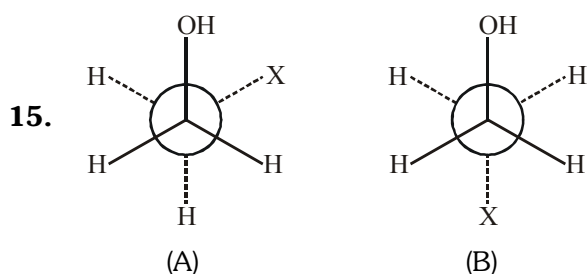
13. The IUPAC name of following structure is:-



- (1) (2E, 4E, 6Z) Octa - 2, 4, 6 - Triene
 (2) (2E, 4E, 6E) Octa - 2, 4, 6 - Triene
 (3) (2Z, 4E, 6Z) Octa - 2, 4, 6 - Triene
 (4) (2Z, 4Z, 6Z) Octa - 2, 4, 6 - Triene

14. Which confirmation of cyclohexane is least stable?

- (1) Chair form (2) Half chair form
 (3) Boat form (4) Twist boat



A is more stable than B, if X is :-

- (1) $-\text{CH}_3$ (2) F
 (3) Br (4) $-\text{C}_2\text{H}_5$

16. Which of the following does not show geometrical isomerism :-

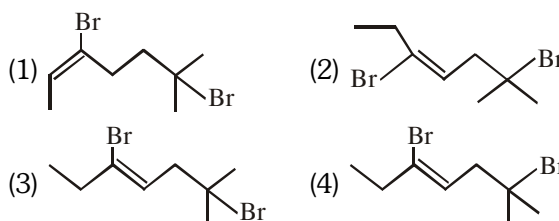
- (1) 1, 2-dichloro- 1-pentene
 (2) 1, 3-dichloro- 2-pentene
 (3) 1, 1-dichloro- 1-pentene
 (4) 1, 4-dichloro- 2-pentene

17. The number of cis-trans isomer possible for the following compound :-

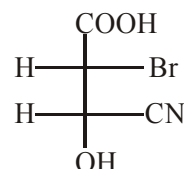


- (1) 2 (2) 4 (3) 6 (4) 8

18. Which of following represent (E)-3,6-dibromo-6-methyl-3-heptene :

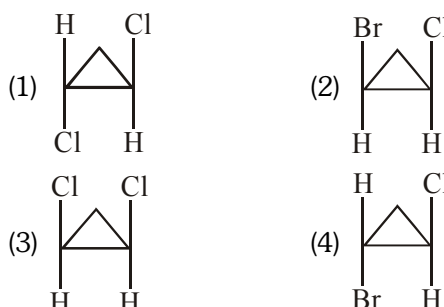


19. The absolute configuration of the two chiral center in the following molecule are -

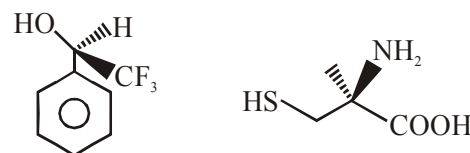


- (1) 2-R, 3-S (2) 2-R, 3-R
 (3) 2-S, 3-S (4) 2-S, 3-R

20. Which of the following compounds is a meso compound :-



21. The R/S configuration of these compounds are respectively :-



- (1) R, R (2) R, S (3) S, R (4) S, S

ANSWER KEY

Exercise-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	4	1	4	1	4	3	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	3	2	2	3	1	2	1	3
Que.	21									
Ans.	2									

EXERCISE-II

- $$\begin{array}{c} \text{H}_3\text{C} \quad \quad \text{H} \\ \quad \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \quad \diagup \quad \diagdown \\ \text{H}_3\text{C} \quad \quad \text{C} \\ \quad \quad \quad | \\ \quad \quad \quad \text{CH}_3 \end{array} \begin{array}{c} \text{H} \\ \diagdown \\ \text{COOH} \end{array}$$

- ed06\BOAI-BO\Kota\JEE(MAIN)\Booster Course Sheet\Chem\Organic Chemistry\02-isomerism.p65

14. Racemic mixture is formed by mixing two-
[AIEEE-2002]

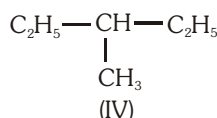
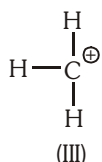
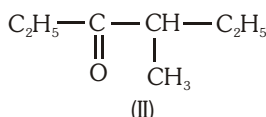
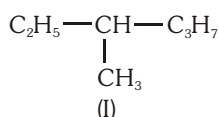
- (1) isomeric compounds
- (2) chiral compounds
- (3) meso compounds
- (4) enantiomers with chiral carbon

15. Geometrical isomerism is not shown by-
[AIEEE-2002]

- (1) 1,1-dichloro-1-pentene
- (2) 1,2-dichloro-1-pentene
- (3) 1,3-dichloro-2-pentene
- (4) 1,4-dichloro-2-pentene

16. Among the following four structures I to IV

[AIEEE-2003]

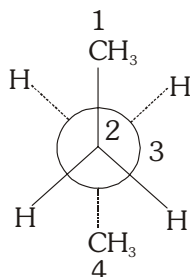


It is true that-

- (1) All four are chiral compounds
- (2) only I and II are chiral compounds
- (3) only III is a chiral compound
- (4) only II and IV are chiral compounds

17. If C_2 in below compound is rotated by 120° angle in anticlockwise direction along C_2-C_3 , which of the following form will be produced -

[IIT-2004]

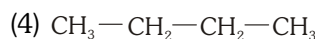
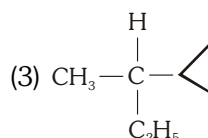
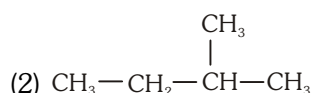


- (1) Staggered
- (2) Perfectly eclipsed
- (3) Perfectly staggered
- (4) Gauche conformation

18. Which of the following will have a meso-isomer also-
[AIEEE-2004]

- (1) 2-chlorobutane
- (2) 2,3-dichlorobutane
- (3) 2,3-dichloropentene
- (4) 2-hydroxy propanoic acid

19. Amongst the following compounds, the optically active alkane having lowest molecular mass is
[AIEEE-2004]



20. Which of following compounds is not chiral
[AIEEE-2005]

- (1) 1-chloropentane
- (2) 2-chloropentane
- (3) 1-chloro-2-methyl pentane
- (4) 3-chloro-2-methyl pentane

21. Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is-
[AIEEE-2005]

- (1) 2-methyl pentane
- (2) 2,2-dimethyl butane
- (3) 2,3-dimethyl butane
- (4) n-hexane

22. Which types of isomerism is shown by 2,3-dichloro butane-
[AIEEE-2005]

- | | |
|----------------|---------------|
| (1) structural | (2) geometric |
| (3) optical | (4) diastereo |

23. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_3 \end{array} \xrightarrow{\text{Cl}_2/h\nu} \text{N (no. of isomers)}$

$\xrightarrow{\text{Fractional distillation}} \text{(F), (N) and (F) are-}$

[IIT-2006]

- | | | | |
|----------|----------|----------|----------|
| (1) 6, 4 | (2) 4, 4 | (3) 6, 6 | (4) 3, 3 |
|----------|----------|----------|----------|

- (4) Isohexane

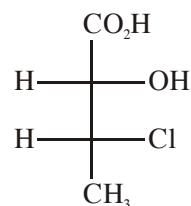
36. How many cyclic structures are possible for C_4H_6 :- [AIEEE-2012(Online)]

- (1) 3 (2) 5
(3) 4 (4) 6

37. Maleic acid and fumaric acids are :- [AIEEE-2012(Online)]

- (1) Tautomers
(2) Chain isomers
(3) Geometrical isomers
(4) Functional isomers

38. The absolute configuration of : [JEE MAINS-2016]



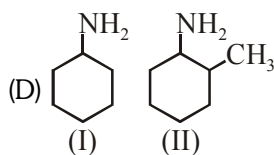
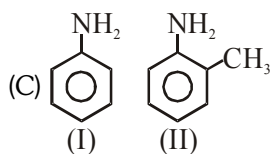
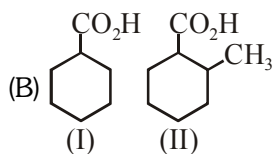
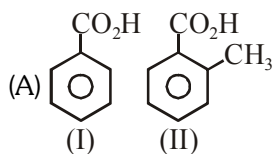
- (1) (2R, 3R) (2) (2R, 3S)
(3) (2S, 3R) (4) (2S, 3S)

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	2	2	2	2	1	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	4	4	1	2	1	2	3	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	1	2	3	1	1	2	1	1
Que.	31	32	33	34	35	36	37	38		
Ans.	2,3,4	4	1	4	3	2	3	3		

GENERAL ORGANIC CHEMISTRY

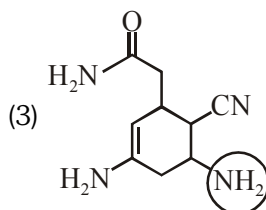
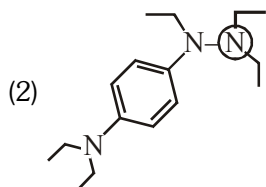
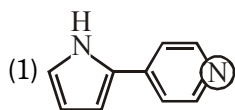
EXERCISE-I

1. In the given pair identify most acidic compound in (A) and (B). Most basic in (C) and (D).

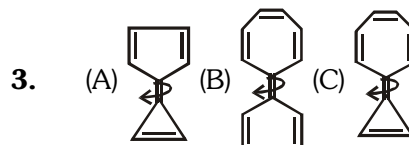


- (1) A-I, B-II, C-I, D-II
(2) A-II, B-I, C-I, D-II
(3) A-II, B-II, C-II, D-II
(4) A-I, B-II, C-I, D-I

2. Circle represent most basic atoms in these molecule. Which of the following is correct representation ?



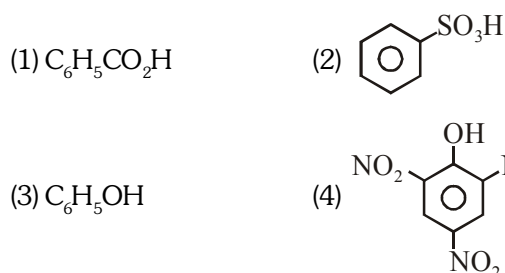
- (4) All of these



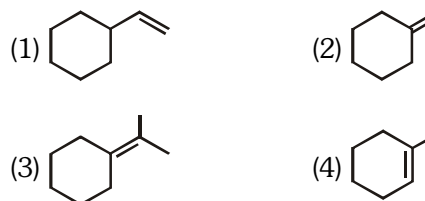
Compare carbon-carbon bond rotation across A, B and C :-

- (1) $A > B > C$ (2) $A > C > B$
(3) $B > A > C$ (4) $B > C > A$

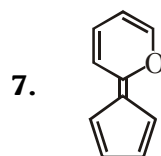
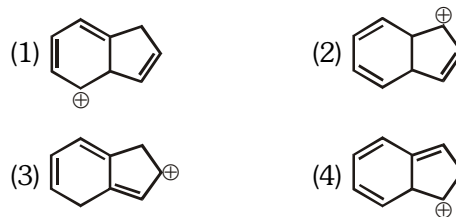
4. One among the following compounds will not give reaction with sodium carbonate :-



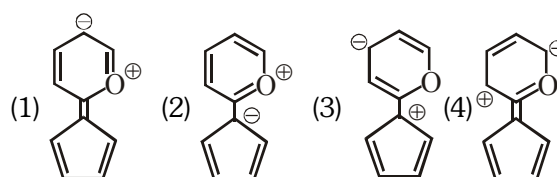
5. Which of the following has the most negative heat of hydrogenation ?



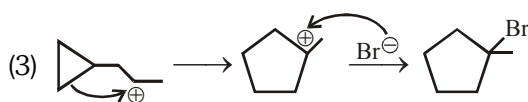
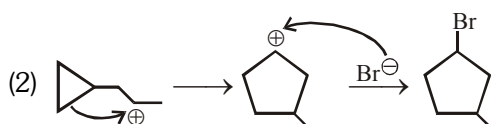
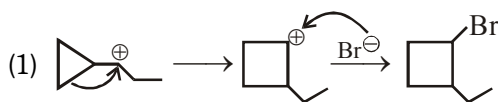
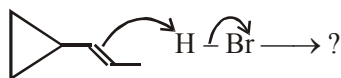
6. Which carbocation is the most stabilized ?



The most stable canonical structure of given molecule is :-

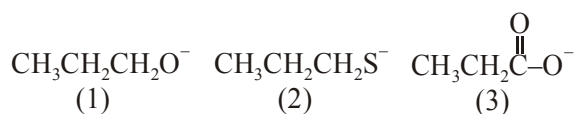


8. Taking into account the stability of various carbocations and, as well as the rules governing mechanisms of carbocation rearrangements, which reaction is most likely to occur during the given reaction :-



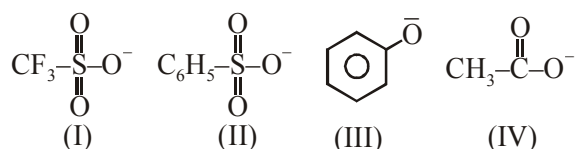
(4) None

9. Rank the following species in order of decreasing nucleophilicity in a polar protic solvent (most→least nucleophilic) :-



- (1) $3 > 1 > 2$ (2) $2 > 3 > 1$
(3) $1 > 3 > 2$ (4) $2 > 1 > 3$

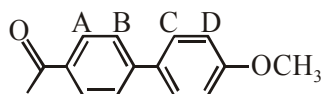
- 10.** Consider the following anions.



When attached to sp^3 -hybridized carbon, their leaving group ability in nucleophilic substitution reaction decreases in the order :-

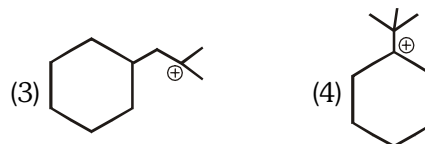
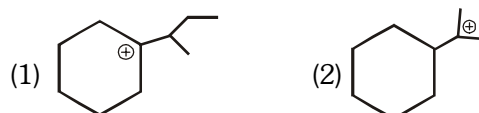
- (1) $I > II > III > IV$
 (2) $I > II > IV > III$
 (3) $IV > I > II > III$
 (4) $IV > III > II > I$

- 11.** Which position will be attacked most rapidly by the nitronium ion ($-\text{NO}_2$)⁺ when the compound undergoes nitration with $\text{HNO}_3/\text{H}_2\text{SO}_4$:-

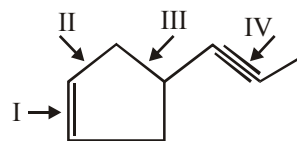


- [illegible]

- 12.** Which cation is most stable :



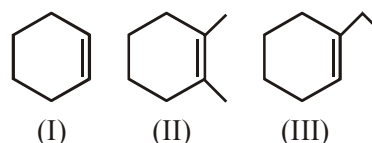
- 13.** What is the correct increasing order of bond lengths of the bonds indicated as I, II, III and IV in following compounds :-



- (1) $I < II < III < IV$ (2) $II < III < IV < I$
(3) $IV < II < III < I$ (4) $IV < I < II < III$

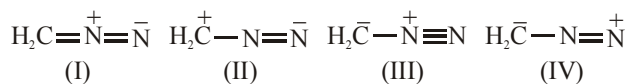
- 14.** Arrange in increasing order of basicity
 CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$, $\text{C}_6\text{H}_5\text{NH}_2$, $(\text{CH}_3)_3\text{N}$
 (1) $(\text{CH}_3)_3\text{N} < (\text{CH}_3)_2\text{NH} < \text{CH}_3\text{NH}_2 < \text{C}_6\text{H}_5\text{NH}_2$
 (2) Reverse of (1)
 (3) $\text{C}_6\text{H}_5\text{NH}_2 < (\text{CH}_3)_3\text{N} < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH}$
 (4) Reverse of (3)

- 15.** Arrange the stability of following :-



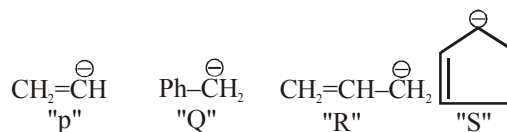
- (1) $I < II < III$ (2) $II < I < III$
(3) $I < III < II$ (4) $II < III < I$

- 16.** The correct stability order of the following resonance structures is :-



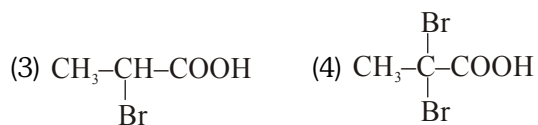
- (1) I > II > IV > III (2) I > III > II > IV
(3) II > I > III > IV (4) III > I > IV > II

- 17.** Arrange the following carbanions in decreasing order of stability :-

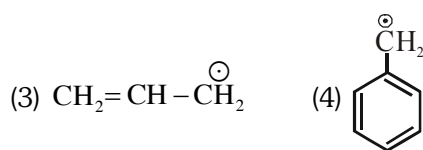


- (1) $P > Q > R > S$
- (2) $S > Q > P > R$
- (3) $S > Q > R > P$
- (4) $Q > S > R > P$

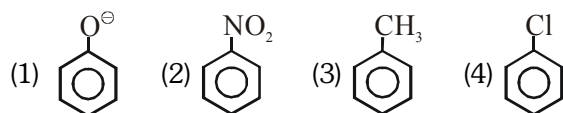
18. Which of the following has highest pK_a value :-
 (1) $\text{Br}-\text{CH}_2-\text{CH}_2-\text{COOH}$ (2) $\text{CH}_3-\text{CH}_2-\text{COOH}$



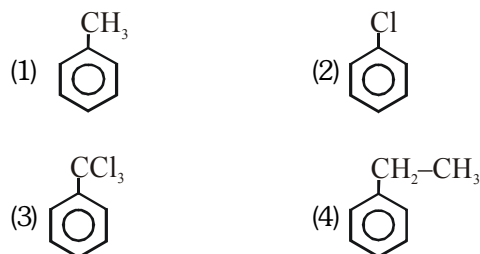
19. Which of the following can show hyperconjugation:-



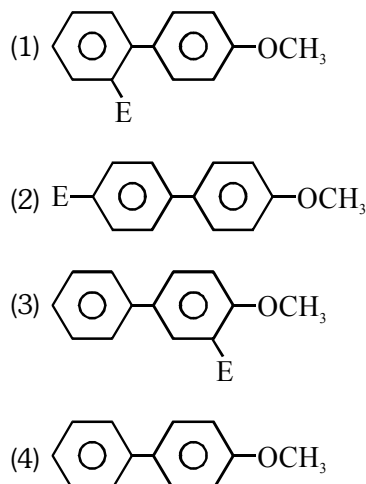
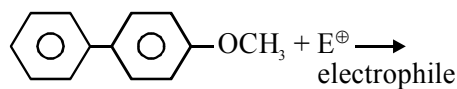
20. Which of the following is most reactive toward electrophilic aromatic substitution reaction :-



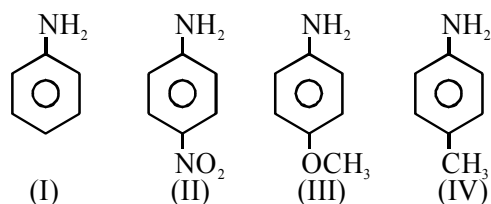
21. In which of the following molecule electrophilic attack at meta positions :-



22. The major product formed in the reaction

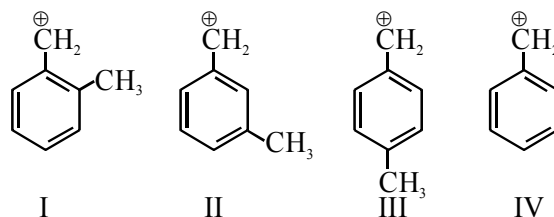


23. Consider the basicity of the following aromatic amines :-



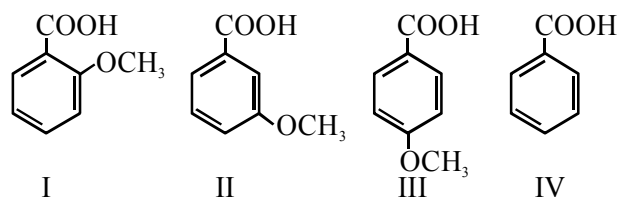
- (1) $\text{II} < \text{I} < \text{IV} < \text{III}$ (2) $\text{I} < \text{II} < \text{III} < \text{IV}$
 (3) $\text{IV} < \text{III} < \text{I} < \text{II}$ (4) $\text{II} < \text{III} < \text{IV} < \text{I}$

24. Arrange the following in decreasing order of stability:-



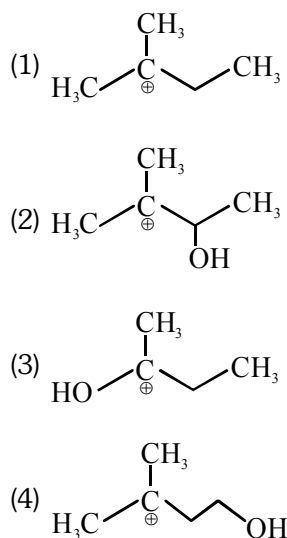
- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{I} > \text{III} > \text{II} > \text{IV}$
 (3) $\text{IV} > \text{III} > \text{II} > \text{I}$ (4) $\text{I} > \text{IV} > \text{II} > \text{III}$

25. Arrange the following in decreasing order of acidic strength :-

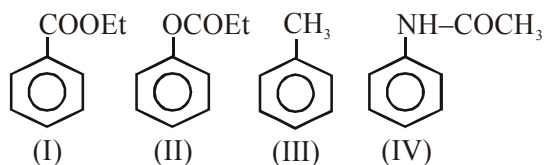


- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{I} > \text{II} > \text{IV} > \text{III}$
 (3) $\text{IV} > \text{III} > \text{II} > \text{I}$ (4) $\text{II} > \text{IV} > \text{III} > \text{I}$

26. Which of the following cation is most stable:-

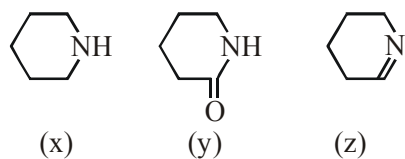


27. Arrange the following in correct order of reactivity for ESR :-



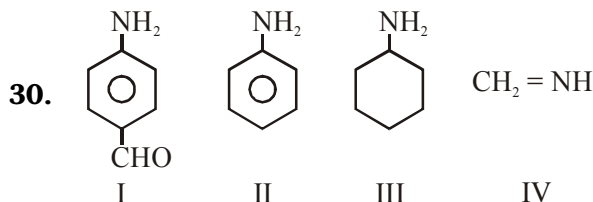
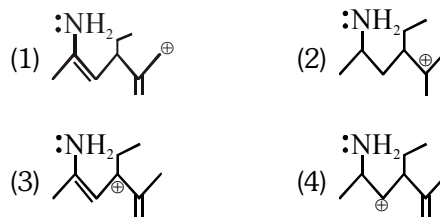
- (1) IV > III > II > I
 (2) I > II > III > IV
 (3) IV > II > I > III
 (4) IV > II > III > I

28. The correct order of basic strengths of x, y, z is :-



- (1) x > y > z (2) x > z > y
 (3) y > x > z (4) y > z > x

29. Among the following which is most stabilised cation:-



Among these compounds the correct order of C-N bond length is :-

- (1) IV > I > II > III (2) III > I > II > IV
 (3) III > II > I > IV (4) III > I > IV > II

ANSWER KEY						Exercise-I				
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	3	3	1	3	2	4	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	3	3	2	3	2	2	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	1	2	2	3	4	2	3	3

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. Which one of the following has the smallest heat of hydrogenation per mole ? [IIT-93]

(1) 1-Butene
(2) trans-2-Butene
(3) cis-2-Butene
(4) 1, 3-Butadiene

2. What is the decreasing order of strength of bases

OH^- , NH_2^- , $\text{HC}\equiv\text{C}^-$, $\text{CH}_3-\text{CH}_2^-$ [IIT-93]

(1) $\text{CH}_3-\text{CH}_2^- > \text{NH}_2^- > \text{HC}\equiv\text{C}^- > \text{OH}^-$

(2) $\text{HC}\equiv\text{C}^- > \text{CH}_3-\text{CH}_2^- > \text{NH}_2^- > \text{OH}^-$

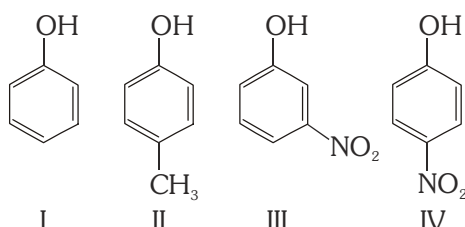
(3) $\text{OH}^- > \text{NH}_2^- > \text{HC}\equiv\text{C}^- > \text{CH}_3-\text{CH}_2^-$

(4) $\text{NH}_2^- > \text{HC}\equiv\text{C}^- > \text{OH}^- > \text{CH}_3-\text{CH}_2^-$

3. The kind of delocalization involving sigma bond orbitals is called [IIT-94]

(1) Inductive effect
(2) Hyperconjugation effect
(3) Electromeric effect
(4) Mesomeric effect

4. In the following compounds [IIT-96]



The order of acidity is -

(1) III > IV > I > II
(2) I > IV > III > II
(3) II > I > III > IV
(4) IV > III > I > II

5. In the following groups : [IIT-97]

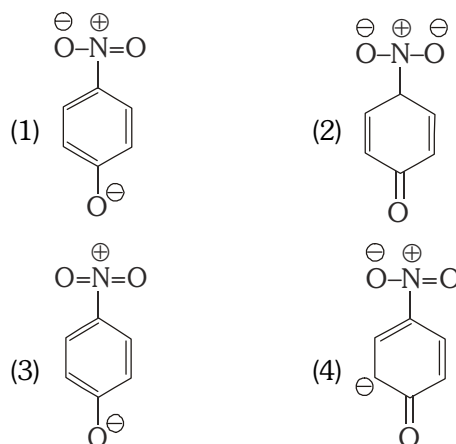
-OAc -OMe -OSO₂Me OSO₂CF₃

I II III IV

the order of leaving group ability is -

(1) I > II > III > IV
(2) IV > III > I > II
(3) III > II > I > IV
(4) II > III > IV > I

6. The most unlikely representation of resonance structure of p-nitrophenoxide ion is - [IIT-99]



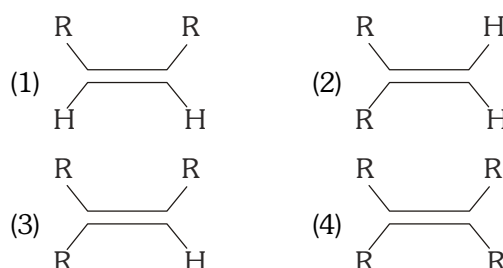
7. An aromatic molecule will not [IIT-99]

(1) have $4n$ π electrons
(2) have $(4n + 2)\pi$ electrons
(3) be planar
(4) be cyclic

8. Amongst the following, the most basic compound is - [IIT-2000]

(1) $\text{C}_6\text{H}_5\text{NH}_2$ (2) $\text{p-NO}_2-\text{C}_6\text{H}_4\text{NH}_2$
(3) $\text{m-NO}_2-\text{C}_6\text{H}_4\text{NH}_2$ (4) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

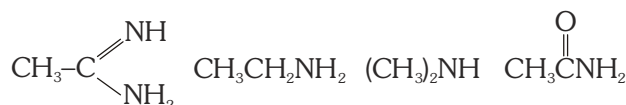
9. Which of the following alkenes will react fastest with H_2 under catalytic hydrogenation conditions [IIT-2001]



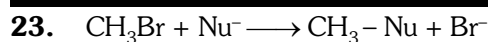
10. Which of the following has the highest nucleophilicity [IIT-2001]

(1) F^- (2) OH^- (3) CH_3^- (4) NH_2^-

11. The correct order of basicities of the following compounds is [IIT-2001]



1 2 3 4
(1) 2 > 1 > 3 > 4 (2) 1 > 3 > 2 > 4
(3) 3 > 1 > 2 > 4 (4) 1 > 2 > 3 > 4



The decreasing order of the rate of the above reaction with nucleophiles (Nu^-) A to D is

[AIEEE-2006]

[$\text{Nu}^- = (\text{A}) \text{PhO}^-$, ($\text{B}) \text{AcO}^-$, ($\text{C}) \text{HO}^-$, ($\text{D}) \text{CH}_3\text{O}^-$]

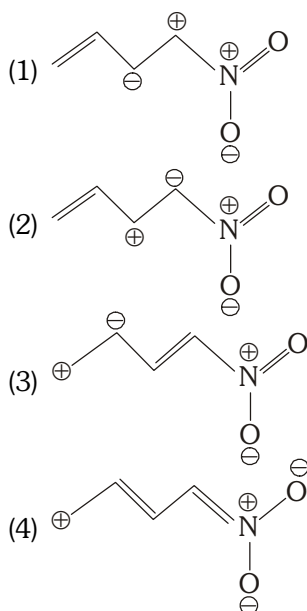
- (1) $\text{D} > \text{C} > \text{B} > \text{A}$ (2) $\text{A} > \text{B} > \text{C} > \text{D}$
(3) $\text{B} > \text{D} > \text{C} > \text{A}$ (4) $\text{D} > \text{C} > \text{A} > \text{B}$

24. The correct order of increasing acid strength of the compounds is [AIEEE-2006]

- (a) $\text{CH}_3\text{CO}_2\text{H}$ (b) $\text{MeOCH}_2\text{CO}_2\text{H}$
(c) $\text{CF}_3\text{CO}_2\text{H}$ (d) $\text{Me}_2\text{C}(\text{Me})\text{CO}_2\text{H}$

- (1) $\text{d} < \text{a} < \text{c} < \text{b}$ (2) $\text{d} < \text{a} < \text{b} < \text{c}$
(3) $\text{a} < \text{d} < \text{c} < \text{b}$ (4) $\text{b} < \text{d} < \text{a} < \text{c}$

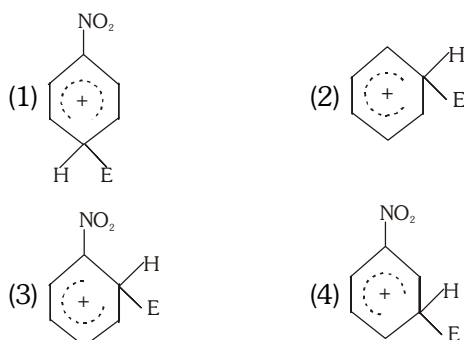
25. Among the following, the least stable resonance structure is - [IIT-2007]



26. Which one of the following is the strongest base in aqueous solution? [AIEEE - 2007]

- (1) Trimethylamine (2) Aniline
(3) Dimethylamine (4) Methylamine

27. The electrophile, E^+ attacks the benzene ring to generate the intermediate σ -complex. Of the following, which σ -complex is of lowest energy? [AIEEE-2008]



28. Arrange the carbanions, $(\text{CH}_3)_3\text{C}^-$, CCl_3^- , $(\text{CH}_3)_2\text{CH}^-$, $\text{C}_6\text{H}_5\text{CH}_2^-$, in order of their decreasing stability :- [AIEEE-2009]

- (1) $\text{CCl}_3^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_2\text{CH}^- > (\text{CH}_3)_3\text{C}^-$
(2) $(\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^- > \text{C}_6\text{H}_5\text{CH}_2^- > \text{CCl}_3^-$
(3) $\text{C}_6\text{H}_5\text{CH}_2^- > \text{CCl}_3^- > (\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^-$
(4) $(\text{CH}_3)_2\text{CH}^- > \text{CCl}_3^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_3\text{C}^-$

29. The correct order of increasing basicity of the given conjugate base ($\text{R}=\text{CH}_3$) is :- [AIEEE-2010]

- (1) $\text{RCOO}^- < \text{HC} \equiv \text{C}^- < \text{NH}_2^- < \text{R}^-$
(2) $\text{RCOO}^- < \text{HC} \equiv \text{C}^- < \text{R}^- < \text{NH}_2^-$
(3) $\text{R}^- < \text{HC} \equiv \text{C}^- < \text{RCOO}^- < \text{NH}_2^-$
(4) $\text{RCOO}^- < \text{NH}_2^- < \text{HC} \equiv \text{C}^- < \text{R}^-$

30. Among the following compounds, the most acidic is [IIT-2011]

- (1) p-nitrophenol
(2) p-hydroxybenzoic acid
(3) o-hydroxybenzoic acid
(4) p-toluic acid

31. The strongest acid amongst the following compounds is ? [AIEEE-2011]

- (1) $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CO}_2\text{H}$
(2) $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{COOH}$
(3) CH_3COOH
(4) HCOOH

32. Ortho-Nitrophenol is less soluble in water than p- and m- Nitrophenols because :- [AIEEE-2012]

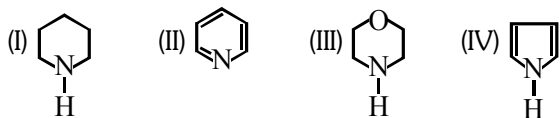
- (1) Melting point of o-Nitrophenol is lower than those of m- and p- isomers
(2) o-Nitrophenol is more volatile in steam than those of m- and p- isomers
(3) o-Nitrophenol shows Intramolecular H-bonding
(4) o-Nitrophenol shows Intermolecular H-bonding

33. Which of the following compounds are antiaromatic :- [AIEEE-2012(Online)]

- (1)
(II)
(III)
(IV)
(V)
(VI)
(1) (III) and (VI) (2) (II) and (V)
(3) (I) and (V) (4) (V) and (VI)

34. In the following compounds :

[AIEEE-2012(Online)]

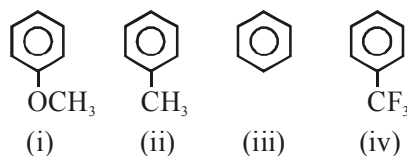


the order of basicity is as follows :

- (1) $IV > III > II > I$ (2) $II > III > I > IV$
 (3) $I > III > II > IV$ (4) $III > I > II > IV$

35. In the below mentioned compounds the decreasing order of reactivity towards electrophilic substitution is : **[AIEEE-2012(Online)]**

[AIEEE-2012(Online)]



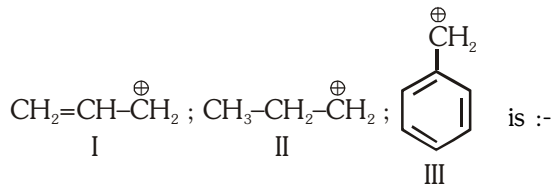
- (1) (iii) > (i) > (iv) > (ii) (2) (iv) > (i) > (ii) > (iii)
(3) (ii) > (iii) > (i) > (iv) (4) (i) > (ii) > (iii) > (iv)

36. The most basic compound among the following is :- **[AIEEE-2012(Online)]**

[AIEEE-2012(Online)]

- (1) Acetanilide (2) Benzylamine
(3) p-Nitro aniline (4) Aniline

37. The order of stability of the following carbocations:

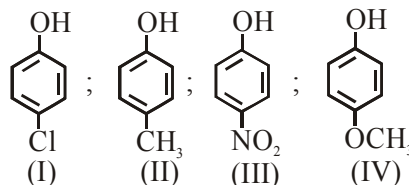


[JEE(MAIN)-2013]

- (1) $III > II > I$
(2) $II > III > I$
(3) $I > II > III$
(4) $III > I > II$

38. Arrange the following compounds in order of decreasing acidity : **[JEE(MAIN)-2013]**

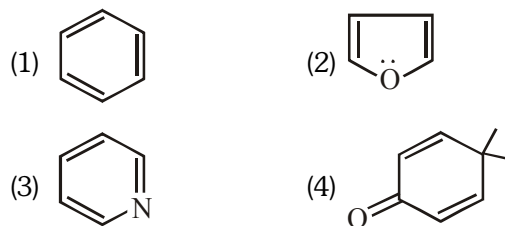
[JEE(MAIN)-2013]



- (1) II > IV > I > III (2) I > II > III > IV
(3) III > I > II > IV (4) IV > III > I > II

39. Which of the following molecules is least resonance stabilized ? **[JEE(MAIN)-2017]**

[JEE(MAIN)-2017]

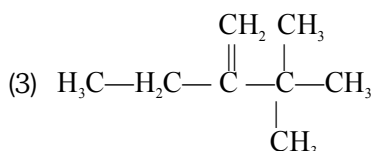
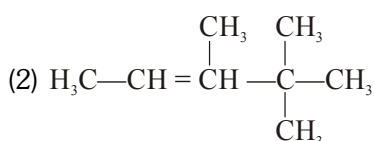
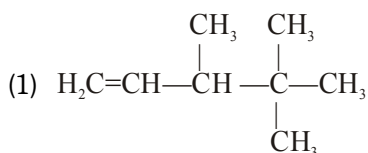
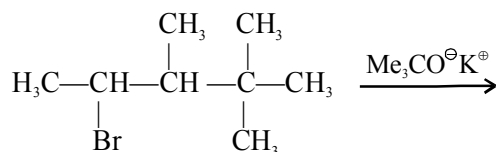


Previous Years Questions				Answer Key			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	2	4	2	3	1	4	1	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	1	3	3	4	4	4	4	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	4	2	1	3	2	1	1	3
Que.	31	32	33	34	35	36	37	38	39	
Ans.	1	3	4	3	4	2	4	3	4	

HYDROCARBON & AROMATIC HYDROCARBON

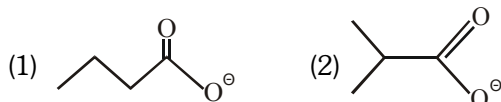
EXERCISE-I

1. What is the major product of the following reaction:



(4) None of these

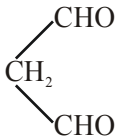
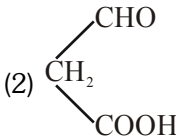
2. Which of the following anion will migrate towards anode to prepare 2,3-Dimethyl butane in Kolbe's electrolysis :-



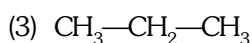
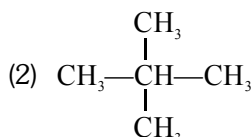
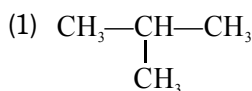
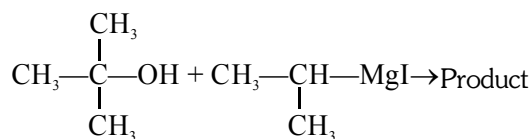
3. Which of the following reagent can not be used to distinguish but-1-yne and but-2-yne :-

- (1) $[\text{Cu}(\text{NH}_3)_2]^\oplus$ (2) NaOH
(3) $[\text{Ag}(\text{NH}_3)_2]^\oplus$ (4) Cold-Dil- KMnO_4

4. $\text{CH}\equiv\text{C}-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OH} \xrightarrow{\text{Dil. H}_2\text{SO}_4/\text{HgSO}_4}$ Major product?

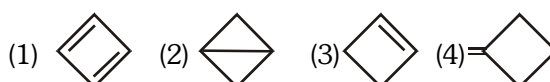
- (1)  (2) 
(3) $\text{CH}\equiv\text{C}-\text{CH}_3$ (4) $\text{CH}_2=\text{CH}-\text{COOH}$

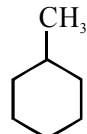
5. What is the product of the following reaction :-



6.  $\xrightarrow{\text{Na/Dry ether}}$ (A) 97%

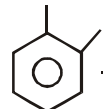
Product (A) of the above reaction is :



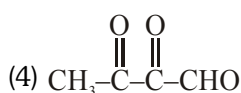
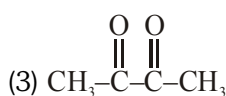
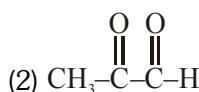
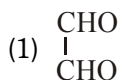
7.  $\xrightarrow{\text{Br}_2/h\nu}$ Monobromo derivatives

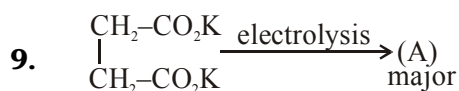
The number of possible monobromo products is (excluding stereoisomers) :

- (1) 4 (2) 5 (3) 8 (4) 10

8.  $\xrightarrow[\text{Zn}]{\text{O}_3}$

Which of the following products cannot be obtained in ozonolysis of o-xylene?



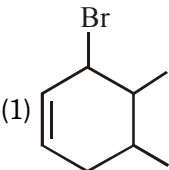
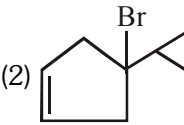
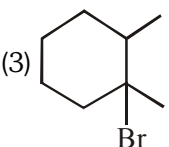
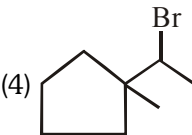


(Kolbe electrolysis method)

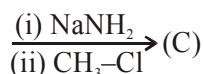
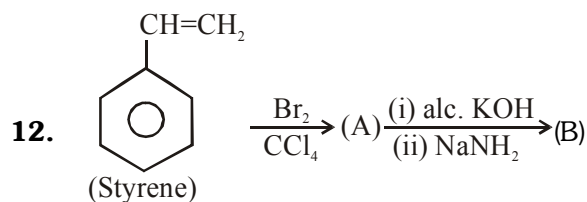
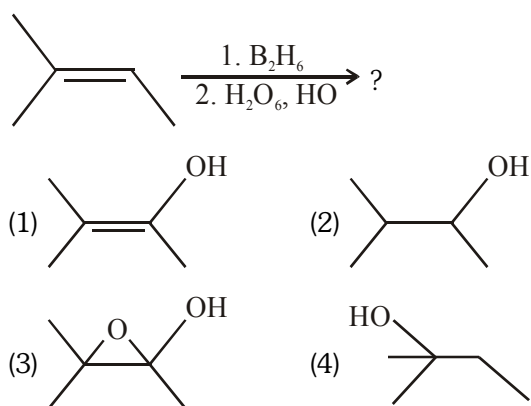
Product (A) of the reaction is :-

- (1) $\text{CH}_3\text{-CH}_3$
- (2) $\text{CH}_2=\text{CH}_2$
- (3) $\text{CH}_3\text{-CH=CH}_2$
- (4) None of these



- (1) 
- (2) 
- (3) 
- (4) 

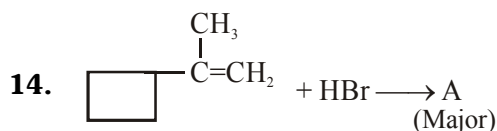
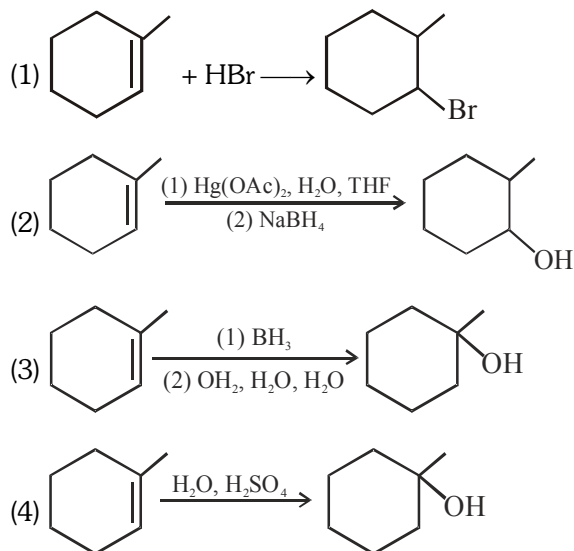
11. The major product of the following reaction sequence is :-



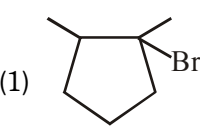
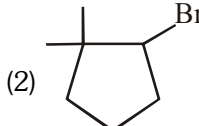
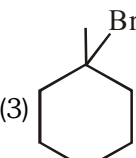
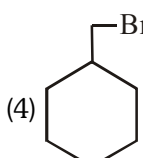
Product (C) is:-

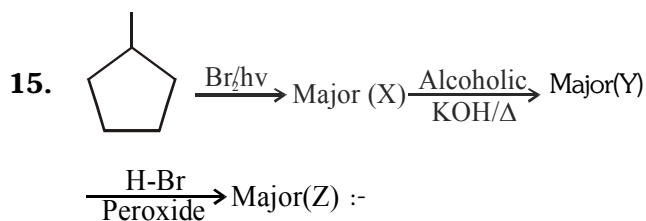
- (1) $\text{Ph-C}\equiv\text{CNa}$
- (2) $\text{Ph-CH}_2\text{-C}\equiv\text{CH}$
- (3) $\text{Ph-C}\equiv\text{C-CH}_3$
- (4) Ph-CH=C=CH_2

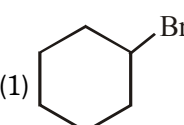
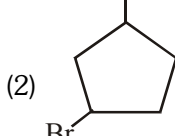
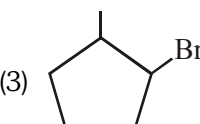
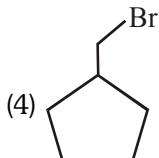
13. Which of the following reactions generates the major product ?

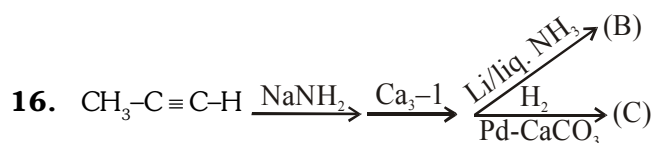


Product (A) is :-

- (1) 
- (2) 
- (3) 
- (4) 



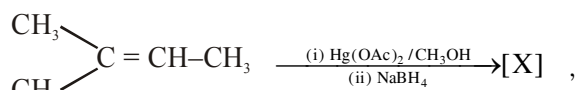
- (1) 
- (2) 
- (3) 
- (4) 



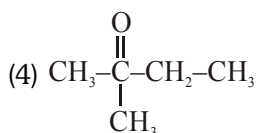
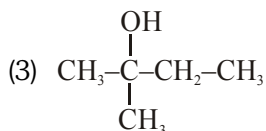
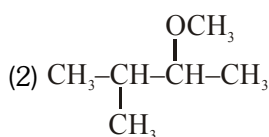
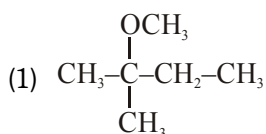
Relation between (B) and (C) is

- (1) Enantiomer (2) Diastereomer
(3) Optical isomers (4) Meso

17. In the given reaction :

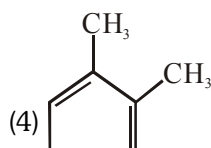
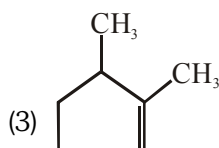
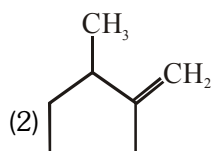
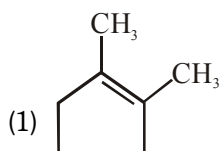
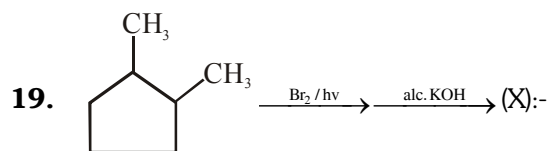


will be :-

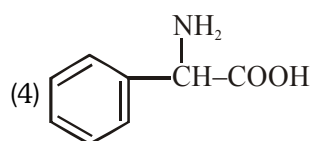
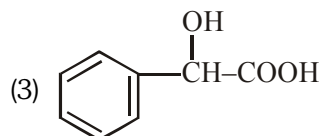
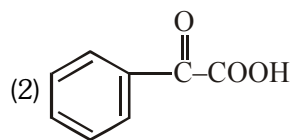
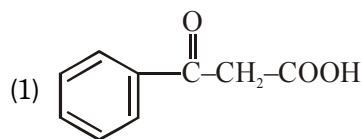


18. Which of the following cannot be considered as a mechanistic step in chain reaction of methane with Cl_2 :-

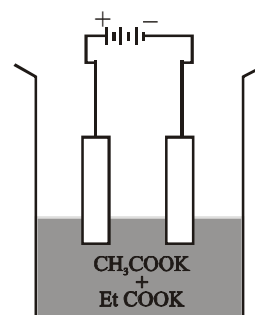
- (1) $\text{Cl}_2 \longrightarrow \text{Cl}^\bullet$
(2) $\text{CH}_4 + \text{Cl}^\bullet \longrightarrow \text{CH}_3\text{Cl} + \text{H}^\bullet$
(3) $\text{Cl}^\bullet + \text{CH}_4 \longrightarrow \text{CH}_3^\bullet + \text{HCl}$
(4) $\text{Cl}^\bullet + \text{CH}_3^\bullet \longrightarrow \text{CH}_3\text{Cl}$



20. Which of the following carboxylic acids undergoes decarboxylation most easily :-



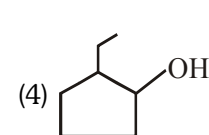
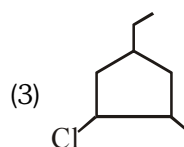
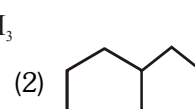
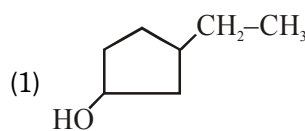
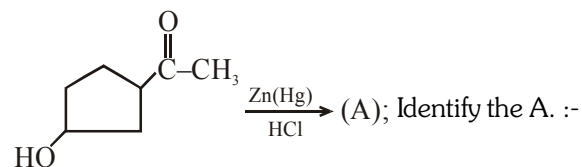
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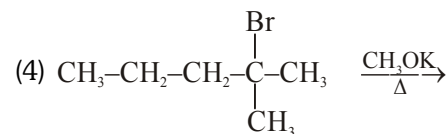
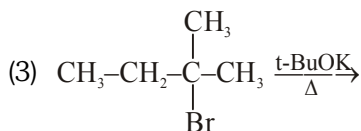
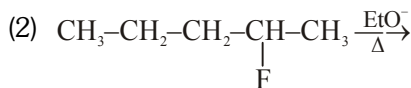
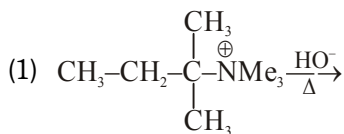
Which of the gases evolved on the surface of anode :-

- (1) $\text{C}_2\text{H}_6, \text{H}_2$
(2) $\text{C}_3\text{H}_8, \text{CO}_2$
(3) $\text{C}_2\text{H}_6 + \text{C}_4\text{H}_6 + \text{CO}_2$
(4) $\text{C}_2\text{H}_6 + \text{C}_4\text{H}_{10} + \text{CO}_2 \uparrow$

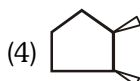
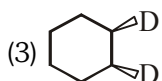
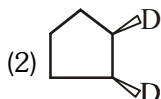
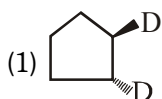
22.



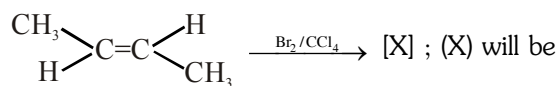
23. In which of the following reaction Saytzeff alkene is major product ?



24.  + D₂ $\xrightarrow{\text{Pd}}$ Major product



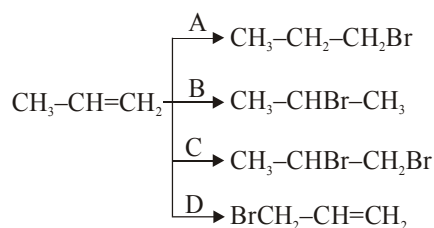
25. In the given reaction



- (1) Meso-2,3-dibromobutane
- (2) Racemic mixture of 2,3-dibromobutane
- (3) Meso as well as racemic mixture
- (4) 1-Bromo-2-butene

26. Match the column :

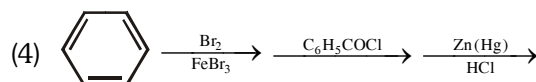
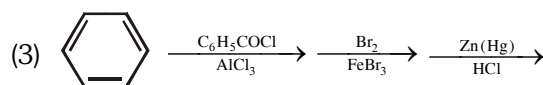
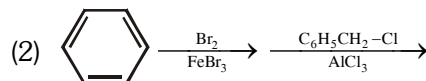
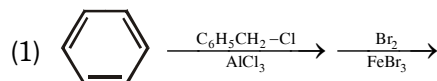
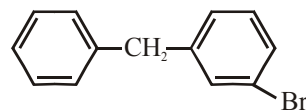
List-I



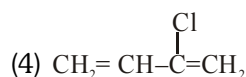
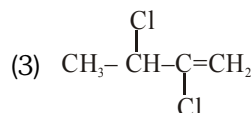
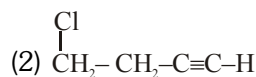
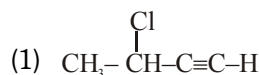
List-II

- | | | | |
|-----------|--|-------|-------|
| (i) HBr | (ii) HBr + peroxide | | |
| (iii) NBS | (iv) Br ₂ , low temperature | | |
| A | B | C | D |
| (1) (ii) | (i) | (iv) | (iii) |
| (2) (i) | (ii) | (iv) | (iii) |
| (3) (i) | (iii) | (iv) | (ii) |
| (4) (ii) | (i) | (iii) | (iv) |

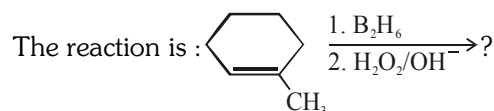
27. Which sequence of steps describes the best synthesis of compound :-



28. $\text{CH}_2=\text{CH-C}\equiv\text{C-H} + \text{H-Cl}$ Product is :-

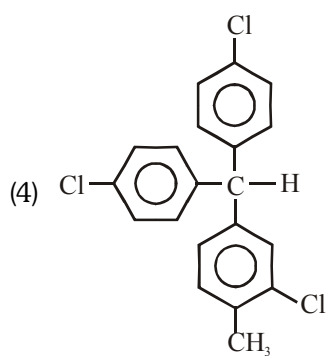
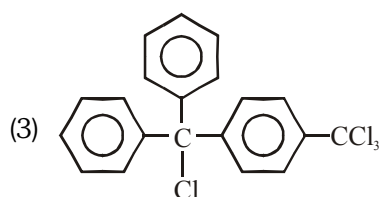
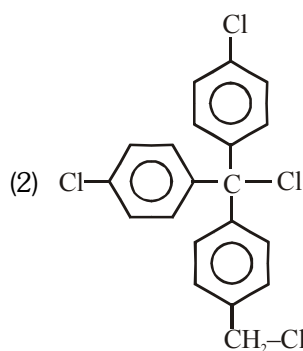
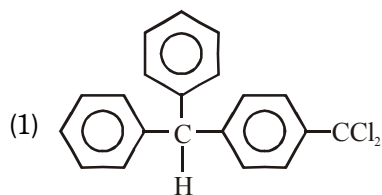
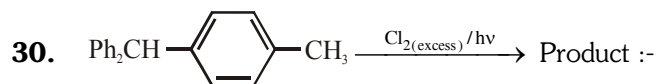


29. 1-Methylcyclohexane is allowed to react with B₂H₆. The product is then treated with H₂O₂ and NaOH.



The product formed is :-

- (1) 1-methyl cyclohexanol
- (2) 2-methyl cyclohexanol
- (3) (±) trans-2-methyl cyclohexanol
- (4) (±) Cis-2-methyl cyclohexanol



			ANSWER KEY				Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	4	2	3	2	2	4	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	4	1	3	2	1	2	1	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	4	2	1	1	3	4	3	3

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. Alcoholic solution of KOH is a specific reagent for [IIT-90]

(1) Dehydration
(2) Dehydrogenation
(3) Dehydro halogenation
(4) Dehalogenation

2. 1-chlorobutane on reaction with alcoholic potash gives [IIT-91]

(1) 1-butene
(2) 1-butanol
(3) 2-butene
(4) 2-butanol

3. The product (s) obtained via oxymercuration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-butyne would be - [IIT-92, 99]

(1) $\text{CH}_3 - \text{CH}_2 - \text{C}(=\text{O}) - \text{CH}_3$
(2) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$
(3) $\text{CH}_3 - \text{CH}_2 - \text{CHO} - \text{HCHO}$
(4) $\text{CH}_3 - \text{CH}_2 - \text{COOH} + \text{HCOOH}$

4. The chief reaction product of reaction between n-butane and bromine at 130°C is - [IIT-95]

(1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ (2) $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$
(3) $\text{CH}_3 - \text{CH}(\text{CH}_3)\text{CH}_2\text{Br}$ (4) $\text{CH}_3 - \text{C}(\text{Br})_2\text{CH}_2\text{CH}_3$

5. When cyclohexane is poured on water, it floats, because - [IIT-97]

(1) Cyclohexane is in 'boat' form
(2) Cyclohexane is in 'chair' form
(3) Cyclohexane is in 'crown' form
(4) Cyclohexane is less dense than water

6. Propyne and propene can be distinguished by - [IIT-2000]

(1) conc. H_2SO_4
(2) Br_2 in CCl_2
(3) dil. KMnO_4
(4) AgNO_3 in ammonia

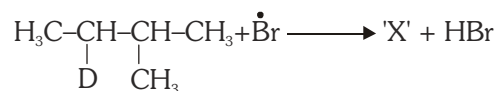
7. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikov addition to alkene because -

(1) Both are highly ionic [IIT Screening-2001]
(2) One is oxidising and the other is reducing
(3) One of the step is endothermic in both the cases
(4) All the steps are exothermic in both cases

8. The reaction of propene with HOCl proceeds via the addition of - [IIT Screening-2001]

(1) H^+ in the first step
(2) Cl^+ in first step
(3) OH^- in first step
(4) Cl^+ and OH^- in single step

9. Consider the following reactions - [IIT Screening-2002]



Identify the structure of the major product 'X'

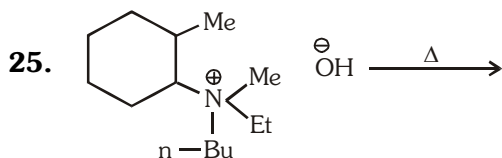
(1) $\text{H}_3\text{C}-\underset{\text{D}}{\text{CH}}-\underset{\text{CH}_3}{\text{CH}}-\dot{\text{C}}\text{H}_2$
(2) $\text{H}_3\text{C}-\underset{\text{D}}{\text{CH}}-\dot{\text{C}}-\text{CH}_3$
(3) $\text{H}_3\text{C}-\dot{\text{C}}-\underset{\text{D}}{\text{CH}}-\text{CH}_3$
(4) $\text{H}_3\text{C}-\dot{\text{C}}\text{H}-\underset{\text{CH}_2\text{D}}{\text{CH}}-\text{CH}_3$

10. Identify a reagent from the following list which can easily distinguish between 1-butyne and 2-butyne [IIT Screening-2002]

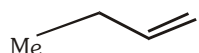
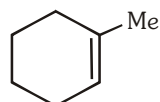
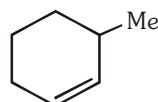
(1) Bromine, CCl_4
(2) H_2 , Lindlar catalyst
(3) dilute H_2SO_4 , HgSO_4
(4) ammonical Cu_2Cl_2 solution

11. Which of these will not react with acetylene - [AIEEE-2002]

(1) NaOH
(2) ammonical AgNO_3
(3) Na
(4) HCl



The alkene formed as a major product in the above elimination reaction is- [AIEEE-06]

- (1)  (2) $\text{CH}_2=\text{CH}_2$
 (3)  (4) 

26. Reaction of trans-2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces- [AIEEE-06]

- (1) 4-phenyl cyclopentene
 (2) 2-phenyl cyclopentene
 (3) 1-phenyl cyclopentene
 (4) 3-phenyl cyclopentene

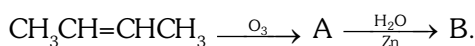
27. Phenyl magnesium bromide reacts with methanol to give- [AIEEE-06]

- (1) A mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$
 (2) A mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
 (3) A mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$
 (4) A mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$

28. Which of the following reactions will yield, 2, 2-dibromopropane [AIEEE-2007]

- (1) $\text{CH}_3-\text{C}\equiv\text{CH} + 2\text{HBr} \longrightarrow$
 (2) $\text{CH}_3\text{CH}=\text{CHBr} + \text{HBr} \longrightarrow$
 (3) $\text{CH}\equiv\text{CH} + 2\text{HBr} \longrightarrow$
 (4) $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{HBr} \longrightarrow$

29. In the following sequence of reactions, the alkene affords the compound 'B' :- [AIEEE-2008]



The compound B is

- (1) $\text{CH}_3\text{CH}_2\text{CHO}$ (2) CH_3COCH_3
 (3) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (4) CH_3CHO

30. The hydrocarbon which can react with sodium in liquid ammonia is [AIEEE-2008]

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_2\text{CH}_3$
 (2) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
 (3) $\text{CH}_3\text{CH}=\text{CHCH}_3$
 (4) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_3$

31. The treatment of CH_3MgX with $\text{CH}_3\text{C}\equiv\text{C}-\text{H}$ produces [AIEEE-2008]

- (1) $\text{CH}_3-\text{CH}=\text{CH}_2$ (2) $\text{CH}_3\text{C}\equiv\text{C}-\text{CH}_3$

- (3) $\text{CH}_3-\text{C}(\text{H})(\text{H})=\text{C}(\text{H})-\text{CH}_3$ (4) CH_4

32. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is :- [AIEEE-2010]

- (1) ethene (2) propene
 (3) 1-butene (4) 2-butene

33. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are - [IIT-2010]

- (1) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
 (2) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$
 (3) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C}\equiv\text{CH}$
 (4) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$

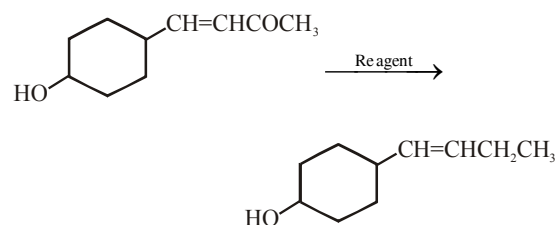
34. Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of :- [AIEEE-2011]

- (1) an isopropyl group
 (2) an acetylenic triple bond
 (3) two ethylenic double bonds
 (4) a vinyl group

35. 2-Hexyne gives trans -2-Hexene on treatment with:- [AIEEE-2012]

- (1) LiAlH_4 (2) Pt/H_2
 (3) Li/NH_3 (4) Pd/BaSO_4

36. In the given transformation, which of the following is the most appropriate reagent ? [AIEEE-2012]

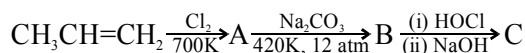


- (1) NaBH_4 (2) $\text{NH}_2\text{NH}_2, \text{OH}^-$
 (3) $\text{Zn} - \text{Hg} / \text{HCl}$ (4) $\text{Na}, \text{Liq. NH}_3$

37. The hydration of propyne results in formation of :- [AIEEE-2012(Online)]

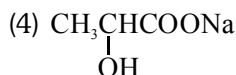
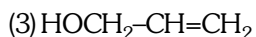
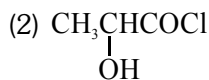
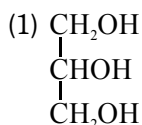
- (1) Propanol-1 (2) Propanal
 (3) Acetone (4) Propene

38. Consider the following sequence of reactions:



compound 'C' is :-

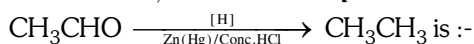
[AIEEE-2012(Online)]



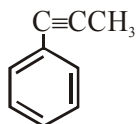
39. The major product obtained in the photobromination of 2-methyl butane is :- [AIEEE-2012(Online)]

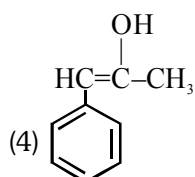
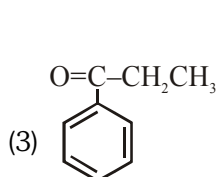
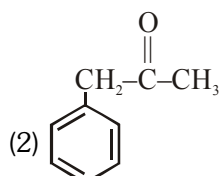
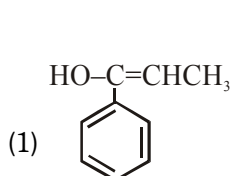
- (1) 2-bromo-3-methylbutane
(2) 1-bromo-2-methylbutane
(3) 1-bromo-3-methylbutane
(4) 2-bromo-2-methylbutane

40. The reaction, [AIEEE-2012(Online)]

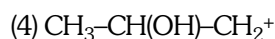
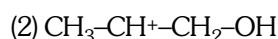


- (1) Cannizaro's reaction
(2) Wolf-Kishner reduction
(3) Rosenmund reduction
(4) Clemmenson reduction

41. In the given reaction  the product 'A' is :- [AIEEE-2012(Online)]



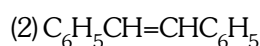
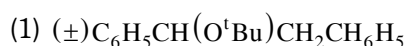
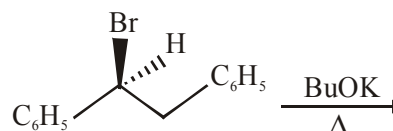
42. The reaction of propene with HOCl ($\text{Cl}_2 + \text{H}_2\text{O}$) proceeds through the intermediate :



43. 3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is :- [JEE MAINS-2017]

- (1) Six (2) Zero
(3) Two (4) Four

44. The major product obtained in the following reaction is :- [JEE MAINS-2017]

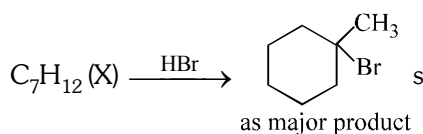


PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	1	2	4	4	3	2	2	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	2	1	1	2	1	3		4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	1	1	2	4	2	1	4	2
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	4	4	4	3	2	3	1	4	4
Que.	41	42	43	44						
Ans.	3	3	4	2						

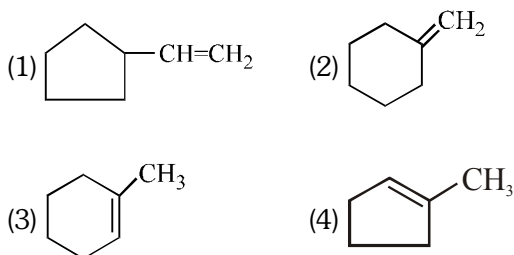
HALOGEN AND ITS DERIVATIVE

EXERCISE-I

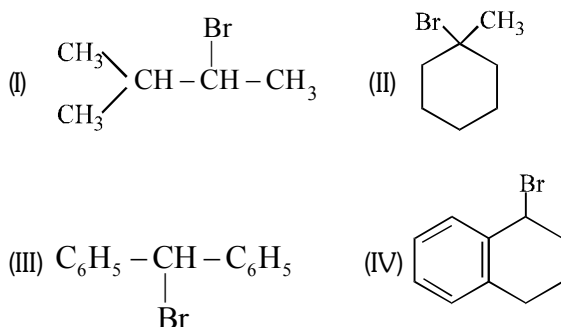
1. In the given reaction



(X) can not be :



2. Consider the S_N1 solvolysis of the following halides in aqueous formic acid:



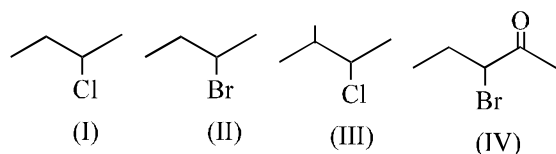
Decide decreasing order of reactivity of above alkyl halide?

- (1) III > IV > II > I (2) II > IV > I > III
(3) I > II > III > IV (4) III > I > II > IV

3. For reaction $\text{CH}_3\text{Br} + \text{OH}^- \longrightarrow \text{CH}_3\text{OH} + \text{Br}^-$ the rate of reaction is given by the expression :

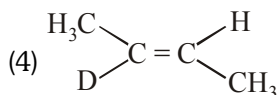
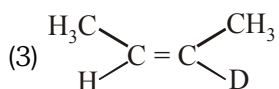
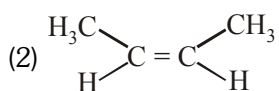
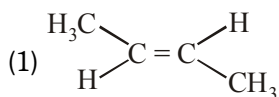
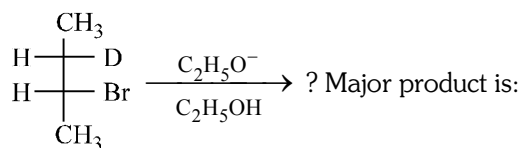
- (1) Rate = $k [\text{CH}_3\text{Br}]$
 (2) Rate = $k [\text{OH}^-]$
 (3) Rate = $k [\text{CH}_3\text{Br}][\text{OH}^-]$
 (4) Rate = $k [\text{CH}_3\text{Br}]^0 [\text{OH}^-]^0$

4. Arrange these compounds in order of increasing S_N2 reaction rate :

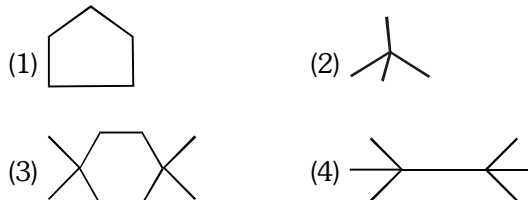


- (1) $\text{III} < \text{I} < \text{II} < \text{IV}$ (2) $\text{III} < \text{II} < \text{I} < \text{IV}$
 (3) $\text{IV} < \text{III} < \text{I} < \text{II}$ (4) $\text{III} < \text{IV} < \text{I} < \text{II}$

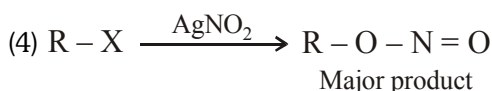
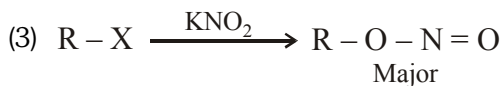
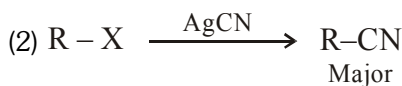
5.



6. Choose that alkane which cannot give only one monochloro derivative upon reaction with chlorine in sun light :



7. Correct option among the following :



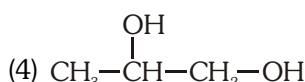
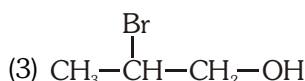
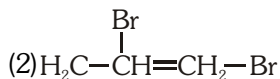
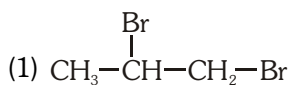
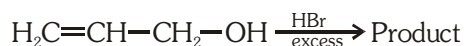
8.



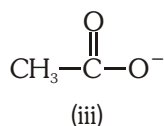
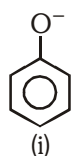
The compound (C) is :-

- (1) CH_3OH (2) HCOOH
(3) CH_3CHO (4) CH_3COOH

17. What is the major product of the following reaction?



18. Arrange the following in order of their nucleophilicity in protic solvent

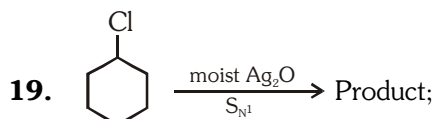


(1) i > ii > iii

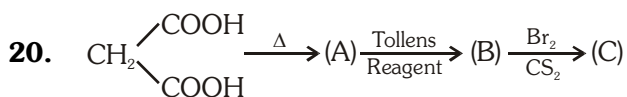
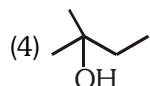
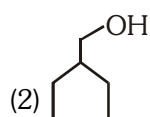
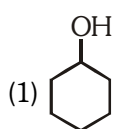
(2) iii > ii > i

(3) ii > i > iii

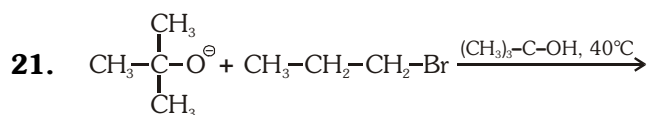
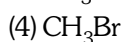
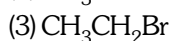
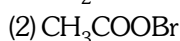
(4) none



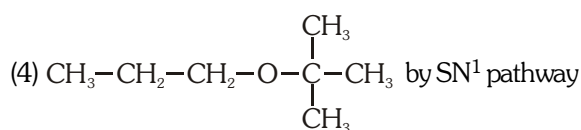
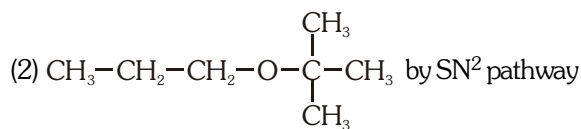
Product of the reaction is :-



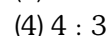
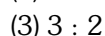
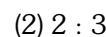
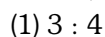
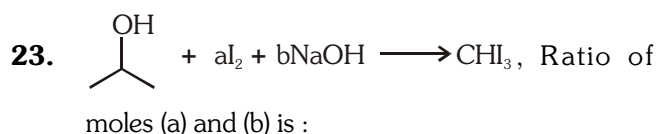
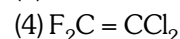
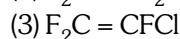
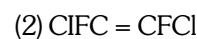
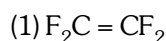
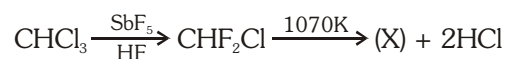
The end product (C) of given sequence of reaction is :-



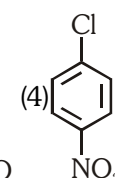
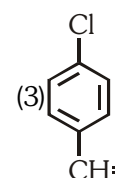
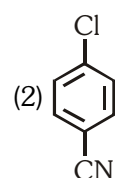
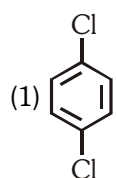
Major product



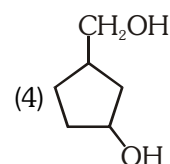
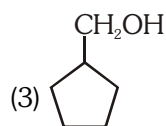
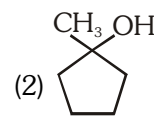
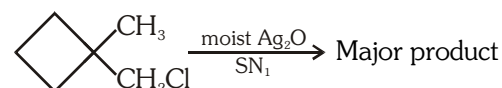
22. Identify (X) :



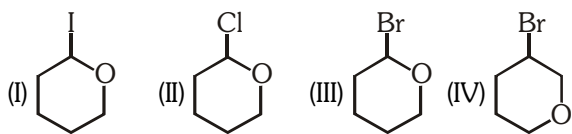
24. Which of the given aryl halide is most readily hydrolysed?



25. Which are possible products in following?

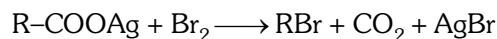


26. Arrange the following compound for S_N^1 reactivity order:

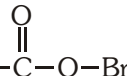
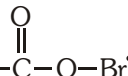


- (1) I > II > III > IV (2) I > III > II > IV
(3) IV > III > II > I (4) II > IV > III > I

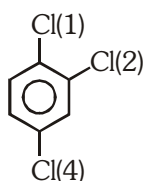
27. In the reaction



The reaction proceeds through the intermediate formation of :

- (1)  (2) 
(3) $R^\bullet$ (4) All of these

28. Which chlorine is most easily replaced under nucleophilic attack?

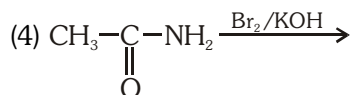
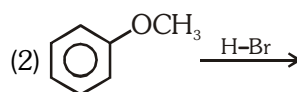
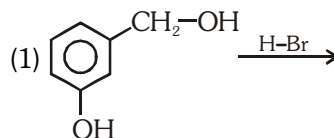


- (1) Cl(1) (2) Cl(2)
(3) Cl(4) (4) Cl(1) and Cl(4)

29. 2-Bromo-1-phenylpropane can be synthesised by :

- (1) $C_6H_5CH_2CH(OH)CH_3 + HBr \xrightarrow{\Delta}$
(2) $C_6H_5CH=CHCH_3 + HBr + \text{benzoyl peroxide} \longrightarrow$
(3) $C_6H_5CH_2CH_2CH_3 + Br_2 + \text{light} \longrightarrow$
(4) none of these

30. In which reaction halide is not obtained ?



				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	3	1	3	3	3	4	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	2	2	2	2	1	3	1	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	2	4	2	2	4	1	2	4

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

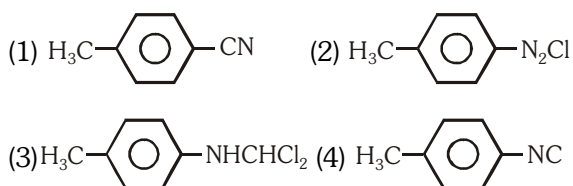
1. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to - **[IIT-90]**
 - (1) The formation of less stable carbonium ion
 - (2) Resonance stabilization
 - (3) The inductive effect
 - (4) sp^3 hybridised carbon attached to the halogen
2. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives : **[IIT 1990]**
 - (1) o-cresol
 - (2) p-cresol
 - (3) 2,4-dihydroxytoluene
 - (4) Benzoic acid
3. 1-Chlorobutane on reaction with alcoholic potash gives - **[IIT-91]**
 - (1) 1-butene
 - (2) 1-butanol
 - (3) 2-butene
 - (4) 2-butanol
4. The products of reaction of alcoholic silver nitrite with ethyl bromide are **[IIT-91]**
 - (1) Ethane
 - (2) Methyl nitrile
 - (3) Nitroethane
 - (4) Ethyl alcohol
5. $(CH_3)_3CMgCl$ reaction with D_2O produces **[IIT-97]**
 - (1) $(CH_3)_3CD$
 - (2) $(CH_3)_3COD$
 - (3) $(CD_3)_3CD$
 - (4) $(CD_3)_3COD$
6. The number of possible enantiomeric pairs that can be produced during monochlorination of isopentane- **[IIT-97]**
 - (1) 2
 - (2) 3
 - (3) 4
 - (4) 1
7. The order of reactivity of the following alkyl halides for a SN^2 reaction is - **[IIT-2000]**
 - (1) $RF > RCl > R-Br > R-I$
 - (2) $R-F > R-Br > R-Cl > R-I$
 - (3) $R-Cl > R-Br > R-F > R-I$
 - (4) $R-I > R-Br > R-Cl > R-F$
8. An S_N2 reaction at an asymmetric carbon of a compound always gives **[IIT-2001]**
 - (1) an enantiomer of the substrate
 - (2) a product with opposite optical rotation
 - (3) a mixture of diastereomers
 - (4) a single stereoisomer

9. Identify the set of reagents/reaction conditions 'X' and 'Y' in the following set of transformations - **[IIT-02]**

$$CH_3-CH_2-CH_2Br \xrightarrow{X} \text{Product} \xrightarrow{Y} CH_3-\underset{\substack{| \\ Br}}{CH}-CH_3$$
 - (1) X = dilute aqueous NaOH, $20^\circ C$, Y = HBr/acetic acid, $20^\circ C$
 - (2) X = concentrated alcoholic NaOH, $80^\circ C$; Y = HBr/acetic acid, 20°
 - (3) X = dilute aqueous NaOH 20° , Y = $Br_2/CHCl_3$, $0^\circ C$
 - (4) X = concentrated alcoholic NaOH, $80^\circ C$; Y = $Br_2/CHCl_3$, $0^\circ C$
10. Following reaction : $(CH_3)_3C-Br + H_2O \longrightarrow (CH_3)_3C-OH + HBr$ is an example of- **[AIEEE-02]**
 - (1) Elimination reaction
 - (2) Free radical substitution
 - (3) Nucleophilic substitution
 - (4) Electrophilic substitution
11. SN^1 reaction is feasible in- **[AIEEE-02]**
 - (1) $\text{Cyclohexyl-Cl} + KOH \longrightarrow$
 - (2) $\text{Cyclopropyl-Cl} + KOH \longrightarrow$
 - (3) $\text{Benzene ring-Cl} + KOH \longrightarrow$
 - (4) $\text{Benzene ring-CH}_2\text{CH}_2\text{-Cl} + KOH \longrightarrow$
12. CH_3MgBr (excess) + Ethyl ester $\longrightarrow$ which can be formed as product **[IIT-03]**
 - (1) $HO-\underset{\substack{| \\ CH_2CH_3}}{C}(CH_2CH_3)_2$
 - (2) $HO-\underset{\substack{| \\ CH_2CH_3}}{C}(CH_3)(CH_2CH_2CH_3)$
 - (3) $HO-\underset{\substack{| \\ CH_3}}{C}(CH_2CH_3)_2$
 - (4) $HO-\underset{\substack{| \\ CH_3}}{C}(CH_3)(CH_2CH_3)$
13. The product of following reaction is **[IIT 2003]**

$$\text{Cyclohexanol} + C_2H_5I \xrightarrow[C_2H_5OH]{C_2H_5O^- (\text{excess})}$$
 - (1) $C_6H_5OC_2H_5$
 - (2) $C_2H_5OC_2H_5$
 - (3) $C_6H_5OC_6H_5$
 - (4) C_6H_5I

14. The reaction of chloroform with alcoholic KOH and p-toluidine form- [AIEEE-03]



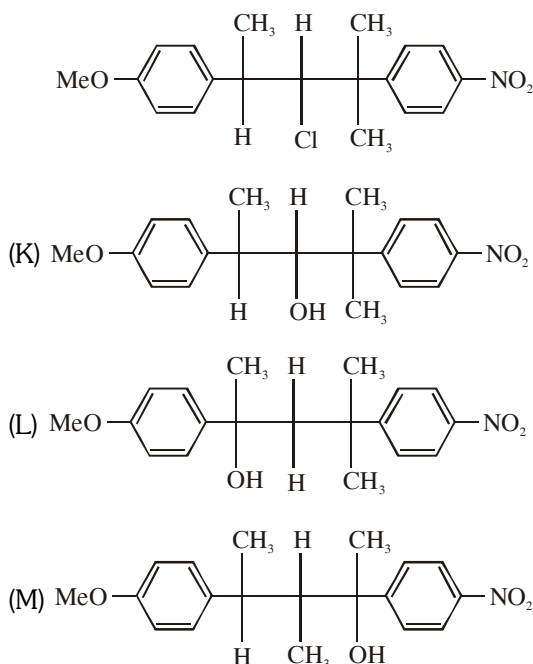
15. The compound formed on heating chlorobenzene with chloral in the presence of concentrated sulphuric acid is- [AIEEE-04]

- (1) Gammaxe (2) DDT
 (3) Freon (4) Hexa chloro ethane

16. Tertiary alkyl halide are practically inert to substitution by S_N2 mechanism because of- [AIEEE-05]

- (1) steric hinderance (2) inductive effect
 (3) instability (4) insolubility

17. The following compound on hydrolysis in aqueous acetone will give: [IIT 2005]

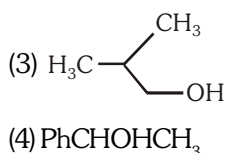


It mainly gives

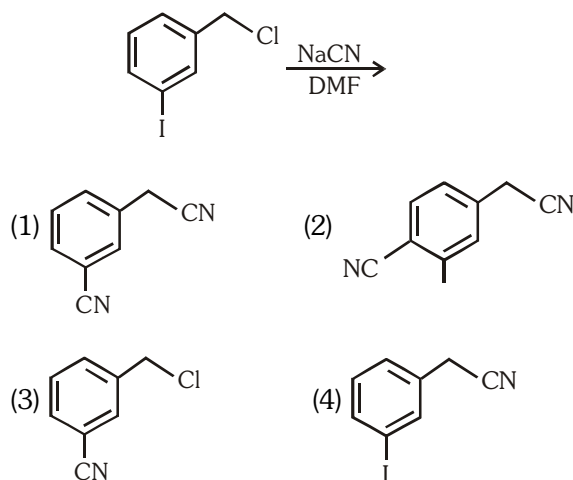
- (1) K and L (2) Only K (3) L and M (4) Only M

18. Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is- [AIEEE-06]

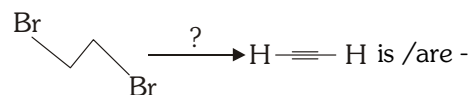
- (1) $CH_3CH_2CH(OH)CH_2CH_3$
 (2) $C_6H_5CH_2CH_2OH$



19. The structure of the major product formed in the following reaction is : [AIEEE-06]



20. The reagent (s) for the following conversion [IIT-07]

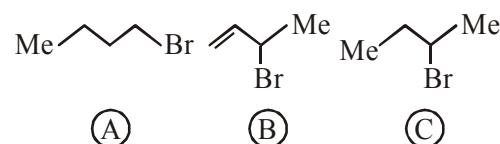


- (1) alcoholic KOH
 (2) Alcoholic KOH followed by $NaNH_2$
 (3) aqueous KOH followed by $NaNH_2$
 (4) Zn/CH_3OH

21. Which of the following on heating with aqueous KOH, produces acetaldehyde ? [AIEEE-09]

- (1) CH_2ClCH_2Cl (2) CH_3CHCl_2
 (3) CH_3COCl (4) CH_3CH_2Cl

22. Consider the following bromides :- [AIEEE-10]



The correct order of S_N1 reactivity is

- (1) $A > B > C$ (2) $B > C > A$
 (3) $B > A > C$ (4) $C > B > A$

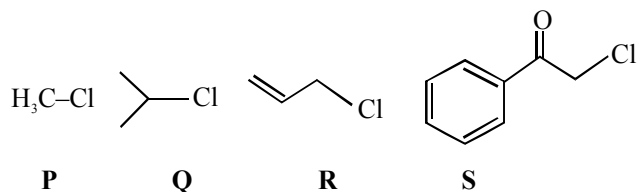
23. Iodoform can be prepared from all except :- [AIEEE-2012]

- (1) Isobutyl alcohol
 (2) Ethyl methyl ketone
 (3) Isopropyl alcohol
 (4) 3-Methyl-2-butanone

24. $C_2H_5Br \xrightarrow{AgCN} X \xrightarrow[Zn-Hg/HCl]{Reduction} Y$, Here, Y is:- [AIEEE-2012(Online)]

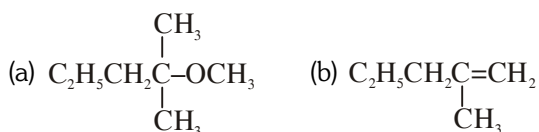
- (1) n-propylamine
 (2) Isopropylamine
 (3) Ethyl methyl amine
 (4) Ethylamine

25. KI in acetone, undergoes S_N2 reaction with each of P, Q, R and S. The rates of the reaction vary as - [IIT-2013]



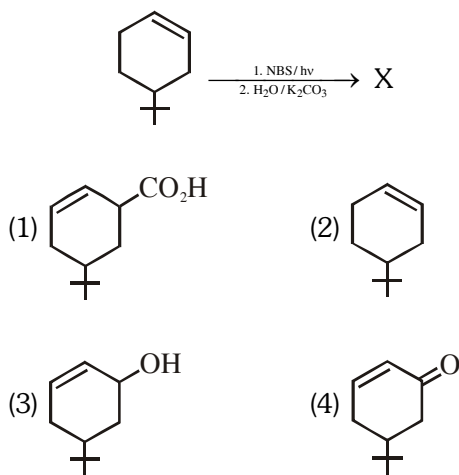
- (1) $P > Q > R > S$ (2) $S > P > R > Q$
 (3) $P > R > Q > S$ (4) $R > P > S > Q$

26. 2-chloro-2-methylpentane on reaction with sodium methoxide in methanol yields : [JEE MAINS 2016]

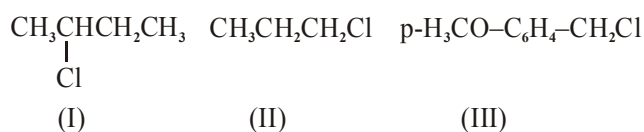


- (1) (a) and (b) (2) All of these
 (3) (a) and (c) (4) (c) only

27. The product of the reaction given below is : [JEE MAINS 2016]

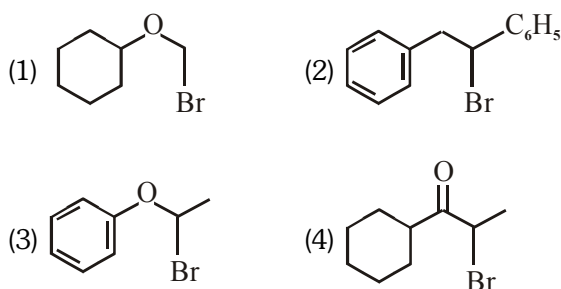


28. The increasing order of the reactivity of the following halides for the S_N1 reaction is [JEE MAINS 2017]

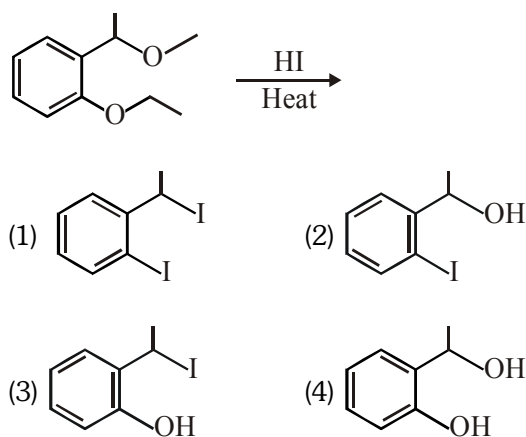


- (1) $(\text{III}) < (\text{II}) < (\text{I})$ (2) $(\text{II}) < (\text{I}) < (\text{III})$
 (3) $(\text{I}) < (\text{III}) < (\text{II})$ (4) $(\text{II}) < (\text{III}) < (\text{I})$

29. Which of the following, upon treatment with tert-BuONa followed by addition of bromine water, fails to decolourize the colour of bromine ? [JEE MAINS 2017]



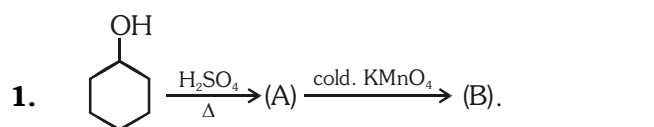
30. The major product formed in the following reaction is : [JEE (Main) 2018]



PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	1	3	1	1	4	2	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	2	4	2	1	1	4	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	1	3	2	2	3	2	1	3

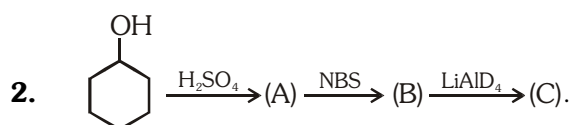
ALCOHOL, PHENOL AND ETHER

EXERCISE-I

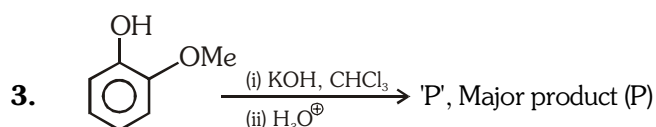
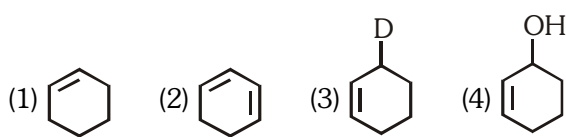


Product (B) is:-

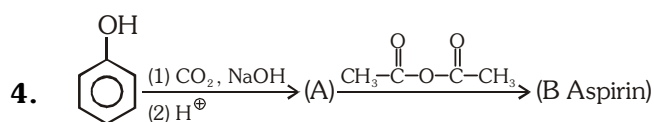
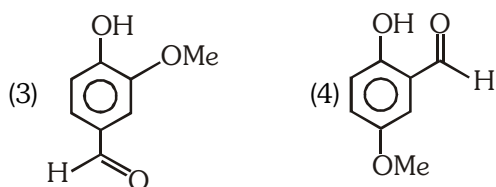
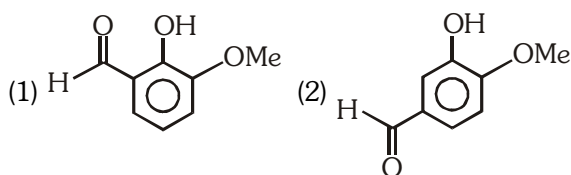
- (1) diastereomer (2) meso
(3) racemic (4) geometrical isomer



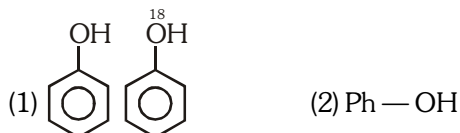
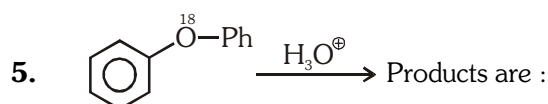
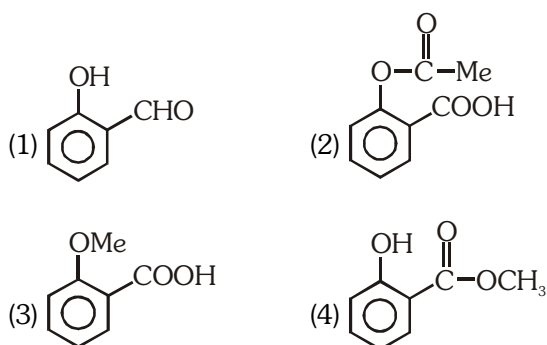
Product (C) is :-



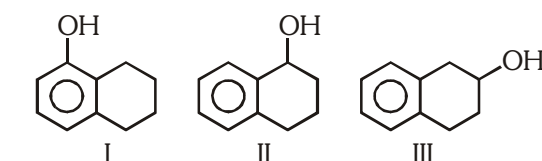
of the reaction is :



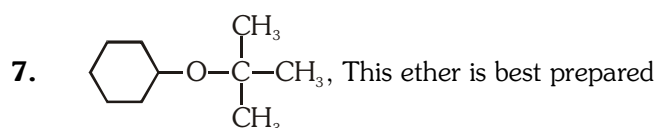
Product (B) is :-



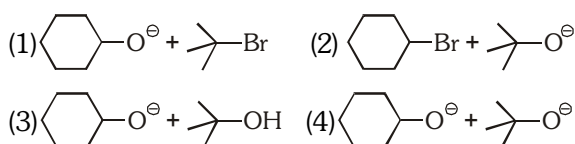
6. Compare the rate of acid catalyst dehydration :



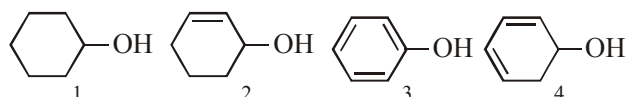
- (1) I > II > III (2) II > III > I
(3) III > II > I (4) II > I > III



by :

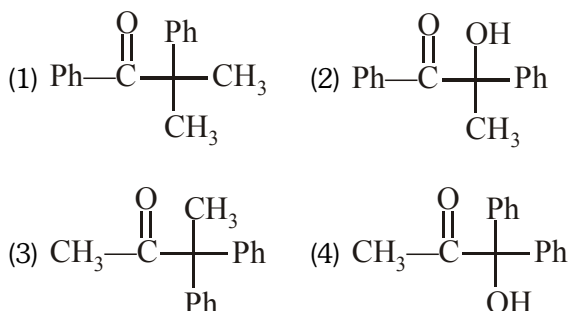
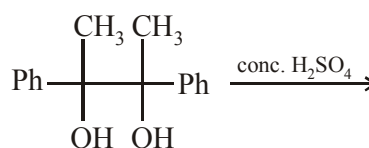


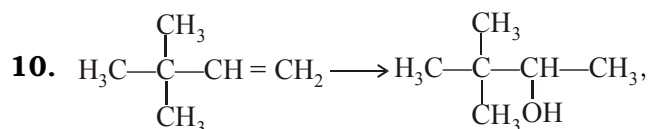
8. Correct order of dehydration of following alcohols will be :



- (1) 1 < 2 < 3 < 4 (2) 4 > 3 > 1 > 2
(3) 4 > 2 > 1 > 3 (4) 1 > 3 > 4 > 2

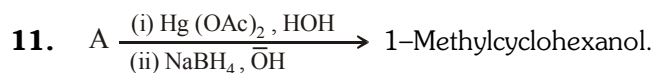
9. Find out major product of following reaction.



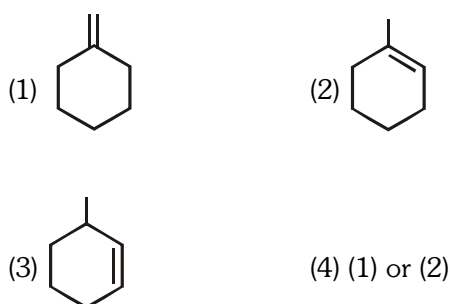


This interconversion can be achieved by :

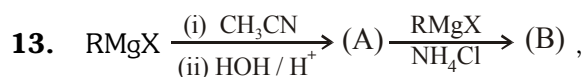
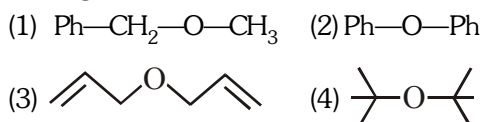
- (1) Acid catalysed hydration
- (2) Oxymercuration-demercuration
- (3) Hydroboration - oxidation
- (4) Any method mentioned above



Here A is :

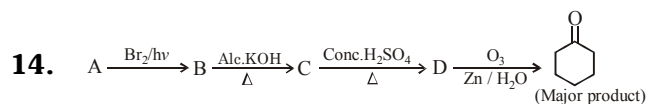


12. Which of the following ethers is least reactive to cleavage with conc. HBr ?

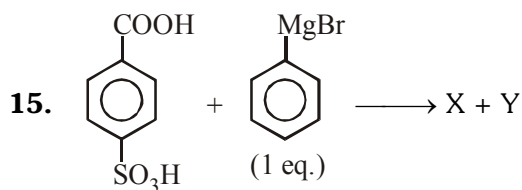
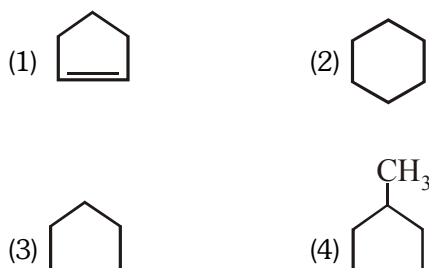


(B) will be :

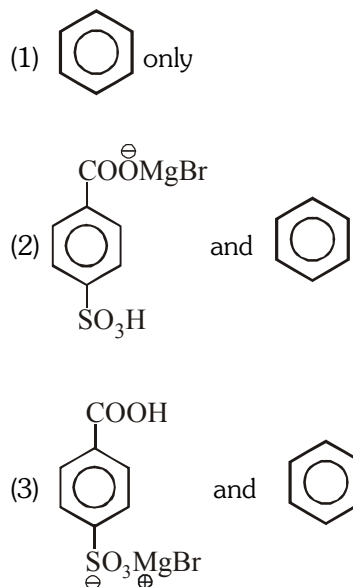
- (1) 1°ROH (2) 2°ROH
- (3) 3°ROH (4) Alkene



Find out the structure of 'A' :

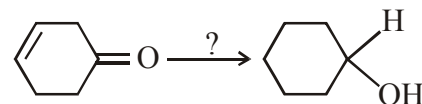


X and Y are respectively :



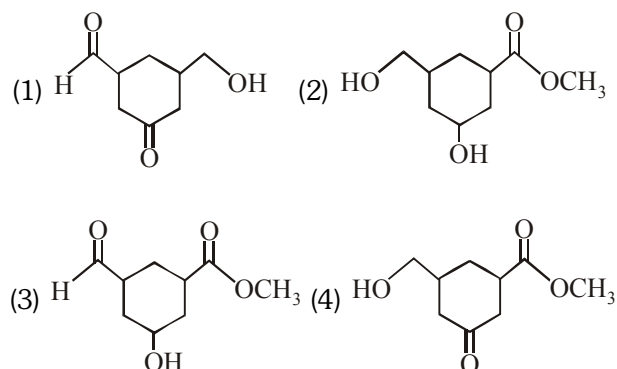
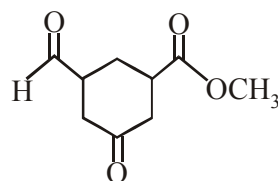
(4) None of these

16. Predict the reducing agents in following reaction.



- (1) LiAlH_4 (2) NaBH_4
- (3) $\text{H}_2 / \text{Pt}, \Delta$ (4) Both (1) and (2)

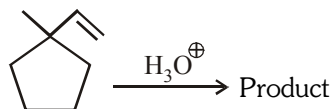
17. Find out the product when the following compound react with NaBH_4 :



18. Compound which does not give alcohol on reduction by LiAlH_4 is/are ?

- (1) $\text{Me}-\text{C}(=\text{O})-\text{Cl}$ (2) $\text{Me}-\text{C}(=\text{O})-\text{NH}_2$
 (3) $\text{Me}-\text{CH}(\text{O})-\text{CH}_2$ (4) $\text{Me}-\text{C}(=\text{O})-\text{O}-\text{C}(=\text{O})-\text{Me}$

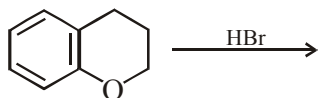
19. In the following reaction



The major product is :

- (1) (2)
 (3) (4)

20. Find out correct product of reaction :

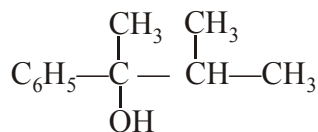


- (1) (2)
 (3) (4)

21. Which of the following alcohol(s) gives immediate turbidity with Lucas reagent?

- (1) (2)
 (3) (4)

22. Which of the following reagents (1 to 4) would you select to convert $\text{C}_6\text{H}_5\text{COCH}_3$ (acetophenone) to the following alcohol ?

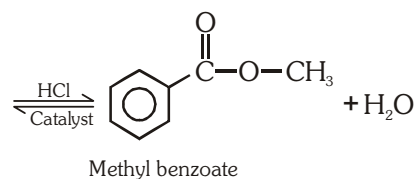


- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$ and hydrolysis
 (2) $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3 \cdot \text{AlCl}_3$
 (3) $(\text{CH}_3)_2\text{CHMgBr}$ and acid hydrolysis
 (4) $\text{CH}_3\text{CHCHCH}_3, \text{Zn}$

23. product is ?

- (1) (2)
 (3) (4)

24. + $\text{CH}_3-\text{O}^{18}-\text{H}$

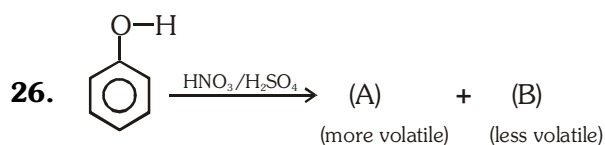


The labelled $-\text{O}^{18}$ will be :-

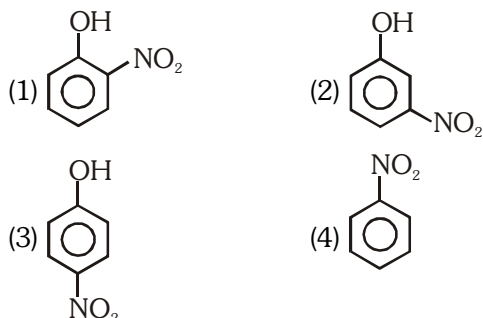
- (1) H_2O (2) Methyl benzoate
 (3) Both (1) and (2) (4) Benzoic acid

25. Which of following compounds are not oxidized by HIO_4 ?

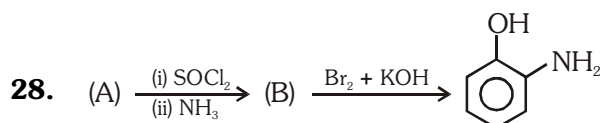
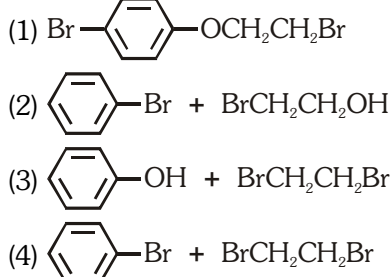
- (1) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_2-\text{OH}$ (2) $\text{CH}_3-\text{C}(=\text{O})-\text{H}$
 (3) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ (4) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{NH}_2$



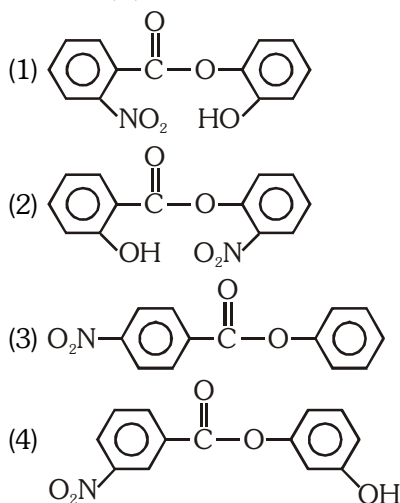
Product (A) of the above reaction is :-



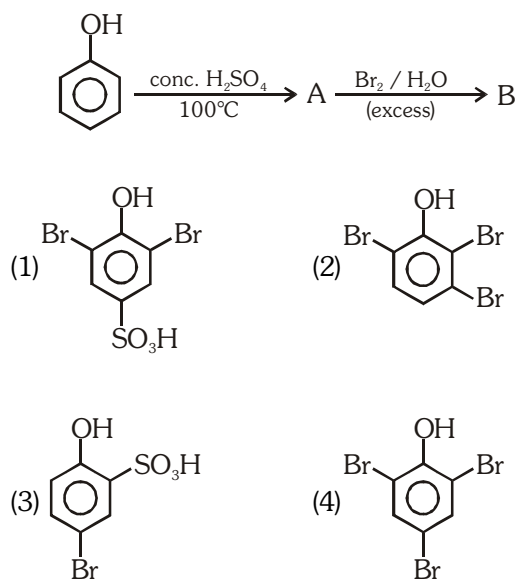
27. What are the products of the following reaction?

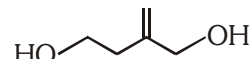


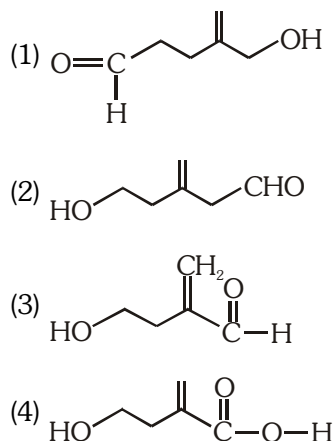
Which of the following compound on hydrolysis gives reactant (A) :-



29. Identify the end product (B) of the following sequence of reactions.



30.  $\xrightarrow{\text{MnO}_2}$ (A), Product (A) is:-



ANSWER KEY

Exercise-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	2	1	2	2	3	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	3	2	3	3	2	2	2	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	3	3	2	4	1	3	2	4	3

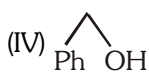
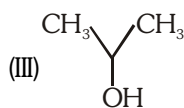
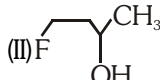
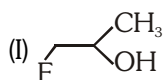
PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. The reaction products of, $C_6H_5OCH_3 + HI \xrightarrow{\Delta}$ are - [IIT-95]



2. The order of reactivity of the following alcohols - [IIT-97]

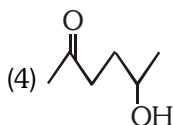
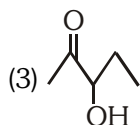
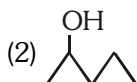


towards conc. HCl is -

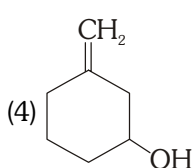
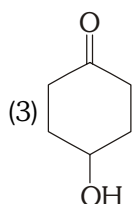
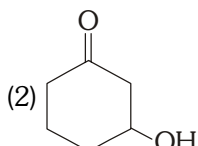
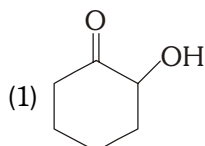
- (1) I > II > III > IV (2) I > III > II > IV
 (3) IV > III > II > I (4) IV > III > I > II
3. Benzenediazonium chloride on reaction with phenol in weakly basic medium gives - [IIT-98]

- (1) Diphenyl ether
 (2) p-hydroxyazobenzene
 (3) Chlorobenzene
 (4) Benzene

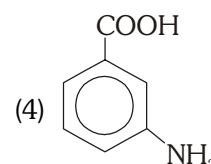
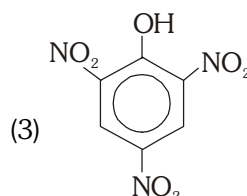
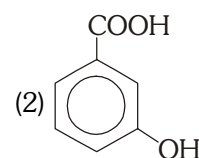
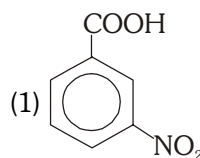
4. Which one of the following will most readily be dehydrated in acidic condition - [IIT-2000]



5. Maximum dehydration takes place that of - [AIIEE-02]



6. Picric acid is - [AIIEE-02]



7. When $CH_2=CH-COOH$ is reduced with $LiAlH_4$, the compound obtained will be - [AIIEE-03]

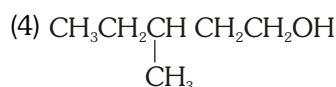
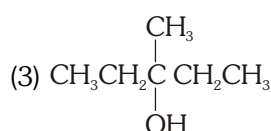
- (1) $CH_3-CH_2-CH_2OH$
 (2) CH_3-CH_2-CHO
 (3) CH_3-CH_2-COOH
 (4) $CH_2=CH-CH_2OH$

8. On acid catalysed hydration, 2-phenyl propene gives : [IIT-04]

- (1) 3-phenyl-2-propanol
 (2) 2-phenyl-1-propanol
 (3) 1-phenyl-3-propanol
 (4) 2-phenyl-2-propanol

9. Among the following compounds which can be dehydrated very easily is - [AIIEE-04]

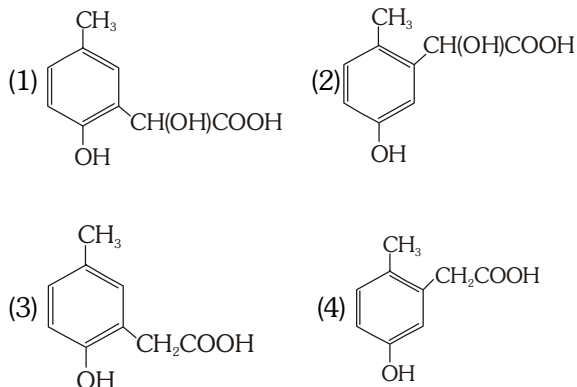
- (1) $CH_3CH_2CH_2CH_2CH_2OH$



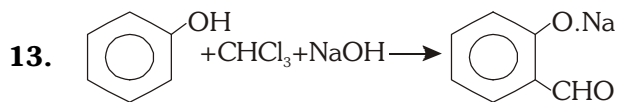
10. Conversion of cyclohexanol into cyclohexene is most effective in [IIT-05]

- (1) concentrated H_3PO_4
 (2) concentrated HCl
 (3) concentrated HCl / $ZnCl_2$
 (4) concentrated HBr

11. p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form, the compound B. The later on acidic hydrolysis gives chiral carboxylic acid. The structure of the carboxylic acid is - [AIEEE-05]



12. HBr reacts with $\text{CH}_2 = \text{CH} - \text{OCH}_3$ under anhydrous conditions at room temperature to give - [AIEEE-06]
- (1) BrCH_2CHO and CH_3OH
 (2) $\text{BrCH}_2 - \text{CH}_2 - \text{OCH}_3$
 (3) $\text{H}_3\text{C} - \text{CHBr} - \text{OCH}_3$
 (4) CH_3CHO and CH_3Br



[AIEEE-06]

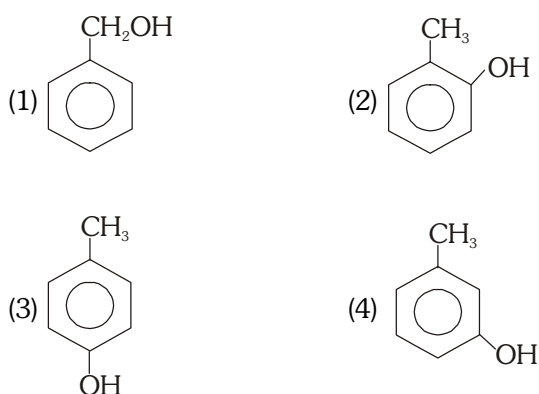
The electrophile involved in the above reaction is -
 (1) dichlorocarbene ($:\text{CCl}_2$)

(2) trichloromethyl anion (CCl_3^-)

(3) formyl cation (CHO^+)

(4) dichloromethyl cation (CHCl_2^+)

14. The structure of the compound that gives tribromo derivative on treatment with bromine water is - [AIEEE-06]



15. In the following sequence of reactions



then compound 'D' is -

[AIEEE-2007]

- (1) Butanal (2) n-Butyl alcohol
 (3) n-Propyl alcohol (4) Propanal

16. A liquid was mixed with ethanol and a drop of concentrated H_2SO_4 was added. A compound with a fruity smell was formed. The liquid was :-

[AIEEE-2009]

- (1) CH_3COCH_3 (2) CH_3COOH
 (3) CH_3OH (4) HCHO

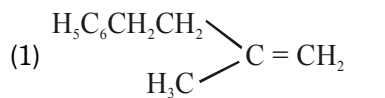
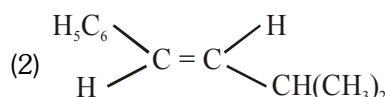
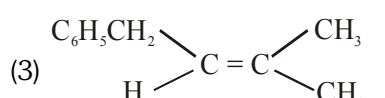
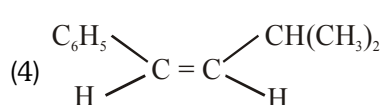
17. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous ZnCl_2 , is :- [AIEEE-10]

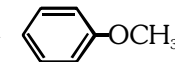
- (1) 1-Butanol
 (2) 2-Butanol
 (3) 2-Methylpropan-2-ol
 (4) 2-Methylpropanol

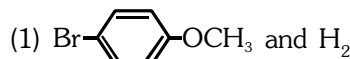
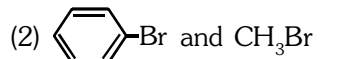
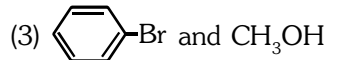
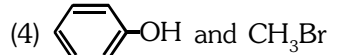
18. The main product of the following reaction is

[AIEEE-10]



- (1) 
- (2) 
- (3) 
- (4) 

19. In the reaction  the products are :- [IIT-10]

- (1)  and H_2
 (2)  and CH_3Br
 (3)  and CH_3OH
 (4)  and CH_3Br

- 20.** Phenol is heated with a solution of mixture of KBr and KBrO_3 . The major product obtained in the above reaction is :- **[AIEEE-11]**

 - (1) 4-Bromophenol
 - (2) 2,4,6-Tribromophenol
 - (3) 2-Bromophenol
 - (4) 3-Bromophenol

21. Which of the following reagents may be used to distinguish between phenol and benzoic acid ? **[AIEEE-11]**

 - (1) Molisch reagent
 - (2) Neutral FeCl_3
 - (3) Aqueous NaOH
 - (4) Tollen's reagent

22. Aspirin can be prepared by the reaction of :- **[AIEEE-2012(Online)]**

 - (1) Salicylic acid with methanol in presence of H_2SO_4
 - (2) Salicylaldehyde with acetic anhydride in presence of H_2SO_4
 - (3) Salicylic acid with acetic anhydride in presence of H_2SO_4
 - (4) Cinnamic acid with acetic anhydride in presence of H_2SO_4

23. Synthetic polymer bakelite can be prepared from following compounds :- **[AIEEE-2012(Online)]**

 - (1) Styrene and vinyl chloride
 - (2) Phenol and formaldehyde
 - (3) Acrylonitrile and vinyl chloride
 - (4) Adipic acid and ethylene glycol

24. Reagent used to convert allyl alcohol to acrolein is :- **[AIEEE-2012(Online)]**

 - (1) MnO_2
 - (2) KMnO_4
 - (3) OsO_4
 - (4) H_2O_2

25. An unknown alcohol is treated with the 'Lucas reagent' to determine whether the alcohol is primary, secondary or tertiary. Which alcohol reacts fastest and by what mechanism :- **[JEE(MAIN)-2013]**

 - (1) secondary alcohol by $\text{SN}1$
 - (2) tertiary alcohol by $\text{SN}1$
 - (3) secondary alcohol by $\text{SN}2$
 - (4) tertiary alcohol by $\text{SN}2$

26. Allyl phenyl ether can be prepared by heating: **[JEE-MAIN-2014]**

 - (1) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Br} + \text{C}_6\text{H}_5\text{ONa}$
 - (2) $\text{C}_6\text{H}_5-\text{CH}=\text{CH}-\text{Br} + \text{CH}_3-\text{ONa}$
 - (3) $\text{C}_6\text{H}_5\text{Br} + \text{CH}_2=\text{CH}-\text{CH}_2-\text{ONa}$
 - (4) $\text{CH}_2=\text{CH}-\text{Br} + \text{C}_6\text{H}_5-\text{CH}_2-\text{ONa}$

27. In the Victor-Meyer's test, the colour given by 1° , 2° and 3° alcohols are respectively :- **[JEE-MAIN-2014]**

 - (1) Red, blue, colourless
 - (2) Colourless, red, blue
 - (3) Red, blue, violet
 - (4) Red, colourless, blue

28. The incorrect combination of names for isomeric alcohols with molecular formula $\text{C}_4\text{H}_{10}\text{O}$ is/are- **[JEE -2014]**

 - (1) *tert*-butanol and 2-methylpropan-2-ol
 - (2) *tert*-butanol and 1, 1-dimethylethan-1-ol
 - (3) *n*-butanol and butan-1-ol
 - (4) isobutyl alcohol and 2-methylpropan-1-ol

29. Williamson synthesis of ether is an example of **[JEE-MAIN-2014]**

 - (1) Nucleophilic addition
 - (2) Electrophilic substitution
 - (3) Nucleophilic substitution
 - (4) Electrophilic addition

30. Bouveault–Blanc reduction reaction involves: **[JEE-MAIN-On-line-2016]**

 - (1) Reduction of an ester with $\text{Na}/\text{C}_2\text{H}_5\text{OH}$
 - (2) Reduction of an ester with H_2/Pd
 - (3) Reduction of a carbonyl compound with Na/Hg and HCl
 - (4) Reduction of an anhydride with LiAlH_4

Previous Years Questions				Answer Key			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	2	1	2	3	4	4	3	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	1	4	3	2	3	2	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	2	1	2	1	1	2	3	1

CARBONYL COMPOUNDS AND CARBOXYLIC ACID & ITS DERIVATIVE

EXERCISE-I

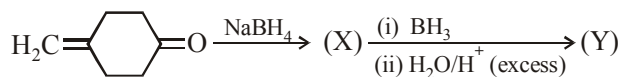
1. Which one of the combinations will give propanaldehyde on dry distillation ?

- (1) $(C_6H_5COO)_2Ca$ and $(HCOO)_2Ca$
- (2) $(CH_3COO)_2Ca$ and $(CH_3CH_2COO)_2Ca$
- (3) $(CH_3-CH_2-COO)_2Ca$ and $(HCOO)_2Ca$
- (4) $(CH_3COO)_2Ca$ and $(CH_3COO)_2Ca$

2. Grignard reagents can never give carbonyl compounds with :

- (1) CO_2 (2) $RCOCl$
- (3) RCN (4) $RCOOR$

3. In the given reaction :



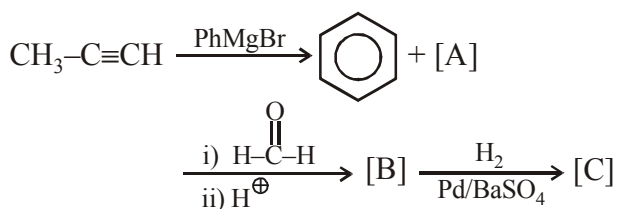
(X) and (Y) are :

- (1) $CH_2=C_6H_{10}OH$ and $HOCH_2-C_6H_{10}O$
- (2) $CH_3-C_6H_{10}O$ and $HOCH_2-C_6H_{10}O$
- (3) $CH_2=C_6H_{10}OH$ and $CH_3-C_6H_{10}OH$
- (4) $CH_2=C_6H_{10}OH$ and $CH_3-C_6H_{10}OH$

4. Acetaldehyde cannot give :

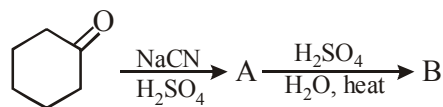
- (1) Iodoform test (2) Lucas test
- (3) Benedict test (4) Tollens test

5. Final product in the given reaction sequence is :



- (1) $CH_3-CH_2-CH_2OH$ (2) $CH_3-CH=CH-CH_2OH$
- (3) $CH_3-CH_2-CH_2OH$ (4) $CH_3-CH=CH-CH_2OH$

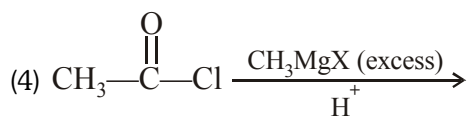
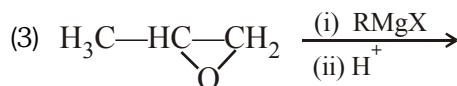
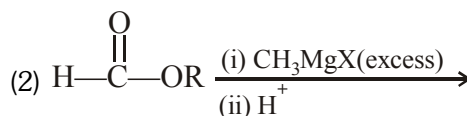
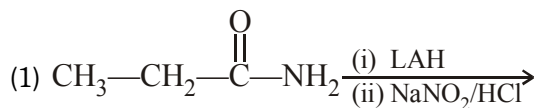
6. Consider the following sequence of reactions.



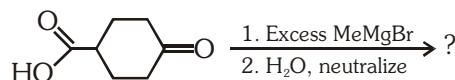
The major product (B) is :

- (1) $C_6H_{10}O$ (2) $C_6H_{10}O$
- (3) $C_6H_{10}O$ (4) $C_6H_{10}O$

7. Which of the following reactions will not give 2° alcohol as a major product :

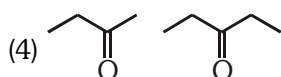
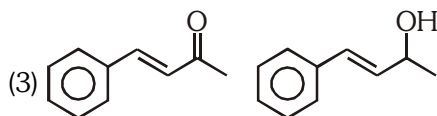
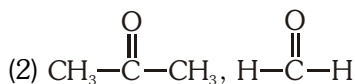
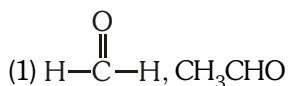


8. Give the expected product of the following reaction.

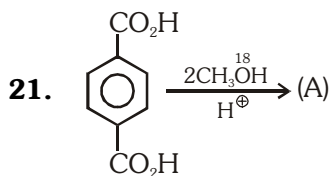
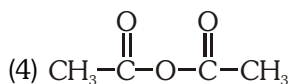
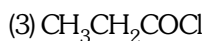
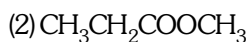


- (1) $HO-C(=O)-C_6H_{10}-C(=O)OH$
- (2) $HO-C(=O)-C_6H_{10}-C(=O)OH$
- (3) $HO-C(=O)-C_6H_{10}-C(=O)OH$
- (4) $HO-C(=O)-C_6H_{10}-C(=O)OH$

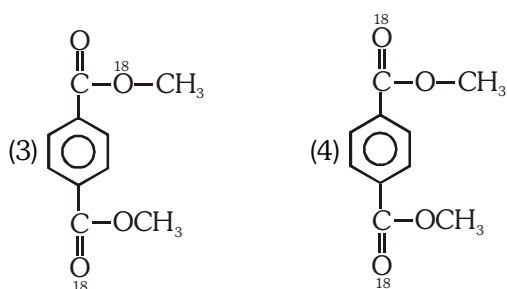
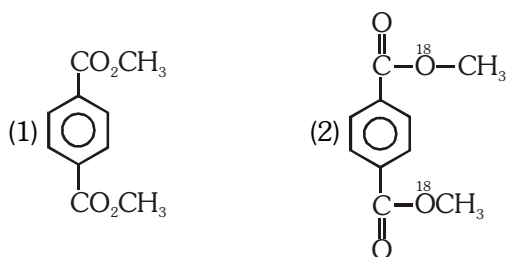
19. Which of the following pair is not differentiated by iodoform test?



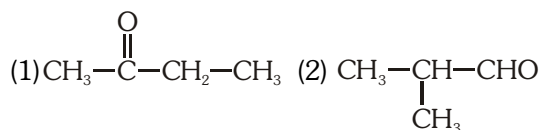
20. The compound which would undergo nucleophilic substitution fastest would be :-



Product (A) is :-

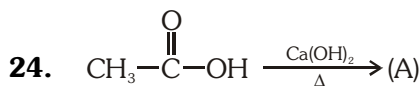
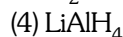
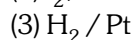
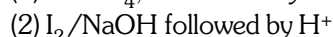
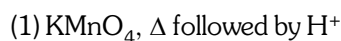


22. Compound (X) $\text{C}_4\text{H}_8\text{O}$, which reacts with 2, 4-DNP derivative and gives negative haloform test is :-

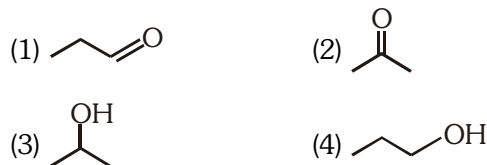


23. $\text{Ph}-\text{CH}=\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \longrightarrow \text{Ph}-\text{CH}=\text{CH}-\text{CO}_2\text{H}$

Above conversion can be achieved by :-

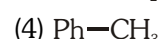
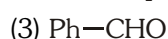
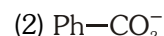
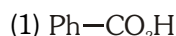


Product (A) is :-



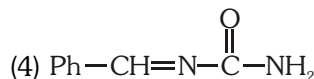
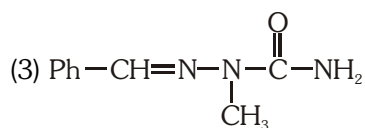
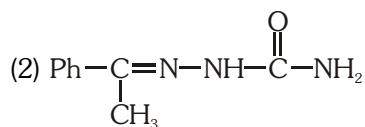
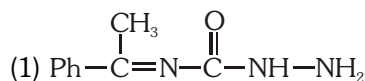
25. $\text{Ph}-\text{CH}_3 \xrightarrow{\text{CrO}_2\text{Cl}_2} (\text{A}) \xrightarrow{\text{conc. KOH}} \text{Ph}-\text{CH}_2\text{OH} + (\text{B})$

Product (B) of above the reaction is :-



26. $\text{Ph}-\underset{\text{OH}}{\text{CH}}-\text{CH}_3 \xrightarrow{\text{PCC}} (\text{A}) \xrightarrow{\text{NH}_2-\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2} (\text{B})$

Product (B) is :-



PREVIOUS YEARS' QUESTIONS

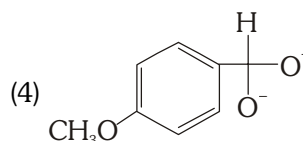
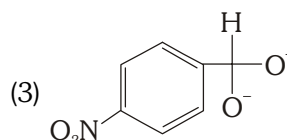
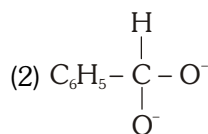
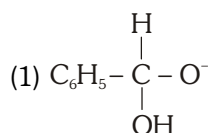
EXERCISE-II

- The formation of cyanohydrin from a ketone is an example of [IIT-90]
 - Electrophilic addition
 - Nucleophilic addition
 - Nucleophilic substitution
 - Electrophilic substitution
- m-chlorobenzaldehyde on reaction with conc. KOH at room temperature gives [IIT-91]
 - Potassium m-chlorobenzoate and m-hydroxybenzaldehyde
 - m-hydroxy benzaldehyde and m-chlorobenzyl alcohol
 - m-chlorobenzyl and m-hydroxybenzyl alcohol
 - Potassium m-chlorobenzoate and m-chlorobenzyl alcohol
- Hydrogenation of benzoyl chloride in the presence of Pd on BaSO₄ gives [IIT-92]
 - Benzyl alcohol
 - Benzaldehyde
 - Benzoic acid
 - Phenol
- An organic compound C₃H₆O does not give a precipitate with 2,4-Dinitrophenyl hydrazine reagent and does not react with metallic sodium compound is - [IIT-93]
 - CH₃CH₂CHO
 - CH₃COCH₃
 - CH₂=CH-CH₂OH
 - CH₂=CH-O-CH₃
- The major product of nitration of Benzoic acid is - [IIT-93]
 - 3-Nitrobenzoic acid
 - 4-Nitrobenzoic acid
 - 2-Nitrobenzoic acid
 - 2,4-dinitrobenzoic acid
- The organic product formed in the reaction

$$\text{C}_6\text{H}_5\text{COOH} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4}$$
 [IIT-95]
 - C₆H₅CH₂OH
 - C₆H₅COOH & CH₄
 - C₆H₅CH₃ & CH₃OH
 - C₆H₅CH₃ & CH₄
- Which of the following carboxylic acids undergo decarboxylation easily [IIT-95]
 - C₆H₅CO-CH₂COOH
 - C₆H₅COCOOH
 - C₆H₅CH₂-COOH
 - C₆H₅CH₂-COOH

- In the Cannizzaro reaction given below

$$2 \text{ Ph-CHO} \xrightarrow{\text{OH}^-} \text{Ph-CH}_2\text{OH} + \text{PhCO}_2^-$$
 the slowest step is- [IIT-96]
 - the attack of OH⁻ at the carbonyl group
 - the transfer of hydride to the carbonyl group
 - the abstraction of proton from the carboxylic acid
 - the deprotonation of Ph-CH₂OH
- Which of the following will give yellow precipitate with I₂/NaOH :- [IIT-97]
 - CH₃CH₂COCH₂CH₃
 - CH₃COOCOCH₃
 - CH₃CONH₂
 - CH₃CH(OH)CH₂CH₃
- In a Cannizzaro reaction the intermediate which is the best hydride donor is - [IIT-97]



- Among the following compounds, which will react with acetone to give a product containing >C=N- [IIT-98]
 - (C₆H₅)₃N
 - (CH₃)₃N
 - C₆H₅NHC₆H₅
 - C₆H₅NHNH₂
- Which of the following will not undergo aldol condensation - [IIT-98]
 - Acetaldehyde
 - Propanaldehyde
 - Benzaldehyde
 - Trideutero acetaldehyde

ALLEN

25. A compound with molecular mass 180 is acylated with CH_3COCl to get a compound with molecular mass 390. The number of amino groups present per molecule of the former compound is :-

[JEE(Main)-2013]

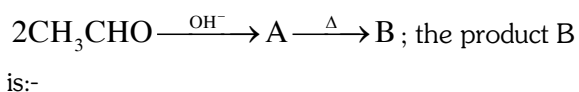
- (1) 2 (2) 5 (3) 4 (4) 6

26. The major organic compound formed by the reaction of 1, 1, 1-trichloroethane with silver powder is :-

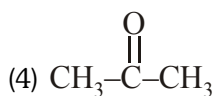
[JEE(Main)-2014]

- (1) 2-Butyne (2) 2-Butene
(3) Acetylene (4) Ethene

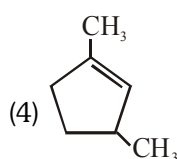
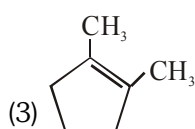
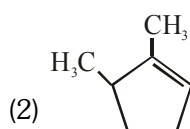
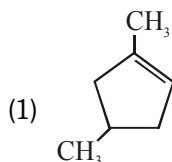
27. In the reaction sequence [JEE(Main)-2015]



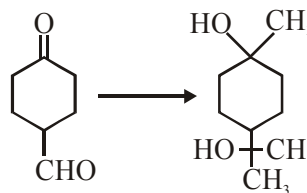
- (1) $\text{CH}_3\text{-CH=CH-CHO}$
(2) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$
(3) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$



28. Which compound would give 5-keto-2-methyl hexanal upon ozonolysis? [JEE(Main) 2015]

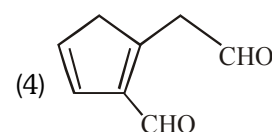
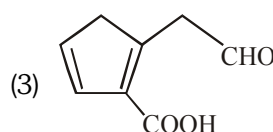
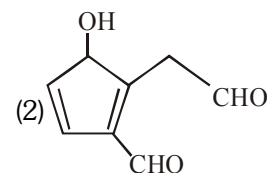
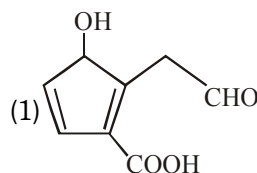
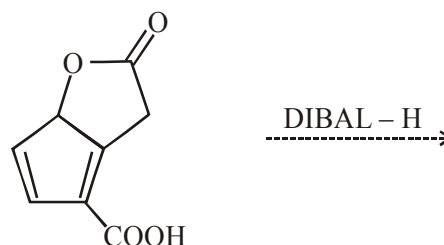


29. The correct sequence of reagents for the following conversion will be :- [JEE(Main)-2017]



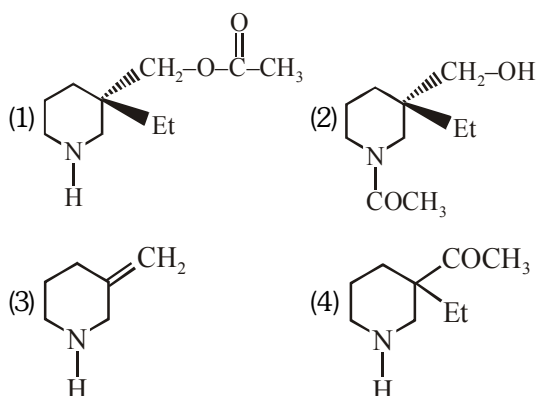
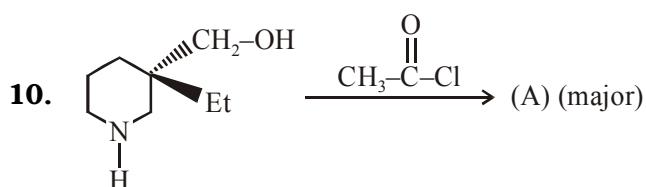
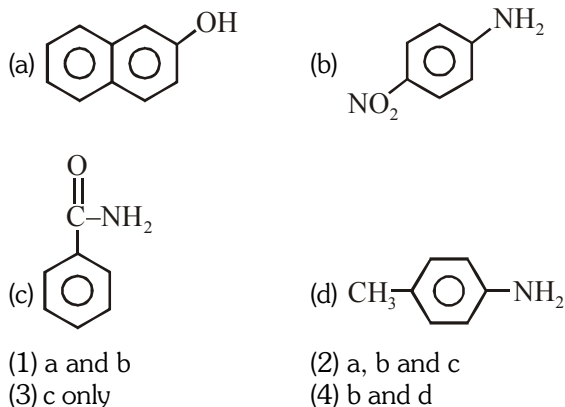
- (1) $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$, $\text{H}^+/\text{CH}_3\text{OH}$, CH_3MgBr
(2) CH_3MgBr , $\text{H}^+/\text{CH}_3\text{OH}$, $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$
(3) CH_3MgBr , $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$, $\text{H}^+/\text{CH}_3\text{OH}$
(4) $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$, CH_3MgBr , $\text{H}^+/\text{CH}_3\text{OH}$

30. The major product obtained in the following reaction is :- [JEE (Main) 2017]

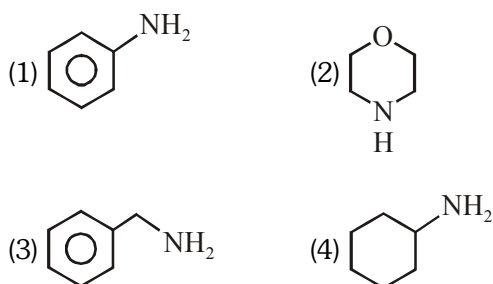


PREVIOUS YEARS QUESTIONS				ANSWER KEY				Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	2	4	2	4	1	1	1	2	4	4	
Que.	11	12	13	14	15	16	17	18	19	20	
Ans.	4	3	1	3	4	1	3	1	2	3	
Que.	21	22	23	24	25	26	27	28	29	30	
Ans.	4	4	3	1	2	1	1	4	1	2	

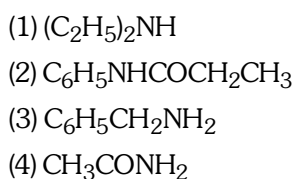
9. Compound X reacts with NaNO_2 & H_2SO_4 & then reacts with ortho cresol to form sharp colour compound, Compound X may be :



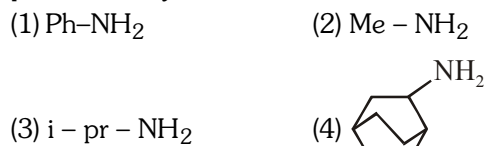
11. Which of the following compounds will not form a Schiff base on reaction with p-nitrobenzaldehyde ?



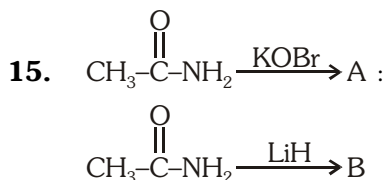
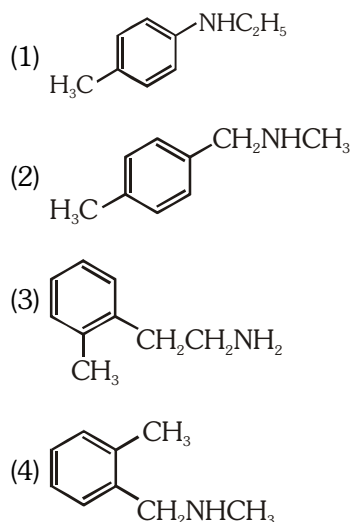
12. Which of the following compounds on treatment with benzene sulphonyl chloride forms an alkali-soluble precipitate ?



13. The amine which cannot be prepared by Gabriel phthalimide synthesis method is :

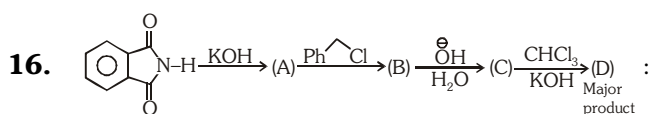


14. Compound A has the formula $\text{C}_9\text{H}_{13}\text{N}$ and forms terephthalic acid with $\text{KMnO}_4/\text{OH}^-$. It forms a compound (in liquid state) with $(\text{COOC}_2\text{H}_5)_2$ which gets decomposed by KOH . The possible structure of compound A is :



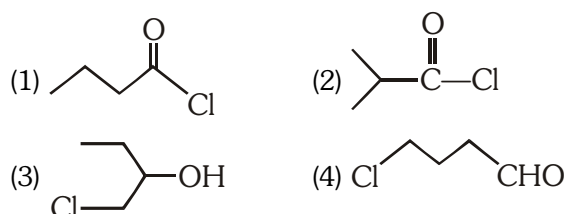
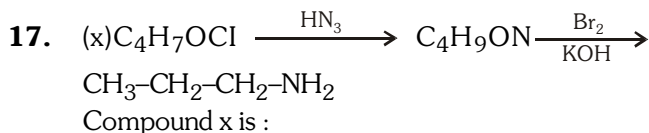
Relation between A and B is

- (1) Chain isomer (2) Homologues
(3) Functional isomer (4) Identical

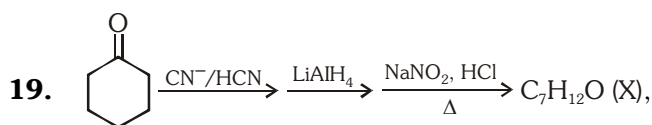
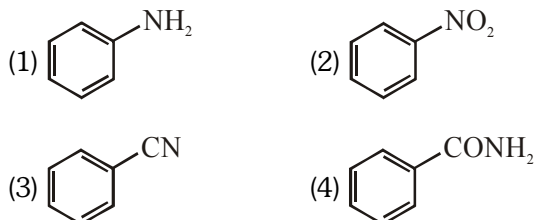


Structure of (D) is :

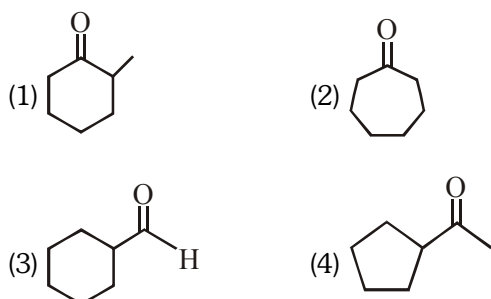
- (1) $\text{Ph-CH}_2\text{NH}_2$ (2) Ph-NC
(3) PhCH_2NC (4) PhCH_2CN



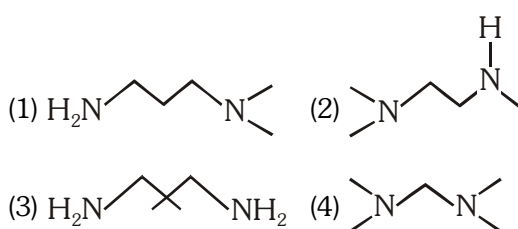
18. A given nitrogen-containing aromatic compound A reacts with Sn/HCl , followed by HNO_2 to give an unstable compound B. B, on treatment with phenol, forms a beautiful coloured compound C with the molecular formula $\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}$. The structure of compound A is :



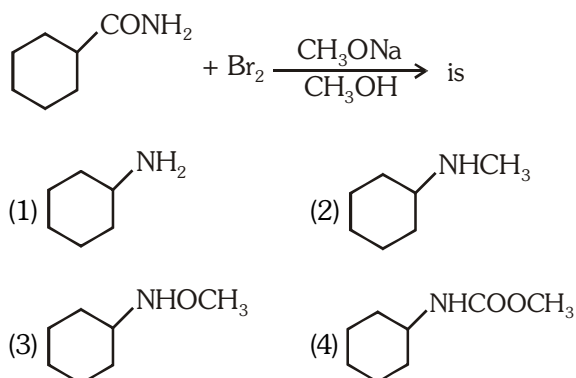
(X) is :



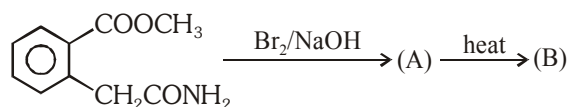
20. The Hinsberg test of a $\text{C}_5\text{H}_{14}\text{N}_2$ compound produces a solid that is insoluble in 10% aq. NaOH . This solid derivative dissolves in 10% aq. H_2SO_4 . Which of the following would best fit these facts ?



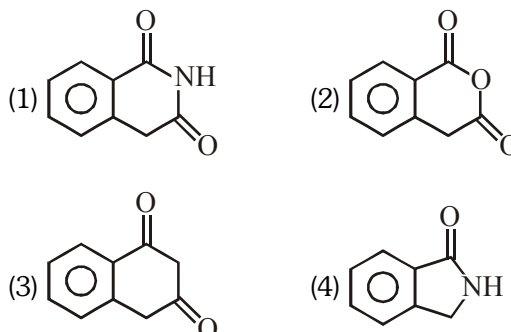
21. The product formed in the reaction



22. The following reactions are carried out.



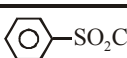
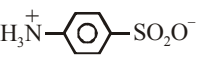
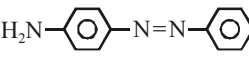
The product (B) is



23. Choose the wrong statement

- (1) 1° Amine gives mustard oil-reaction
 (2) 1° Amine forms salt with H_2PtCl_6
 (3) 1° Amine gives hydrogen gas with sodium
 (4) 1° Amine gives alcohol by hydrolysis

24. Match the reagents given in Column-I to the structures given in Column-II

Column-I		Column-II	
(a)	Zwitter ion structure	(p)	
(b)	Schiff's base	(q)	$\text{C}_6\text{H}_5\text{N}(\text{CH}_3)\text{NO}$
(c)	Hinsberg's Reagent	(r)	
(d)	Nitroso amine	(s)	
(e)	Azodye	(t)	$\text{R}-\text{CH}=\text{NR}$

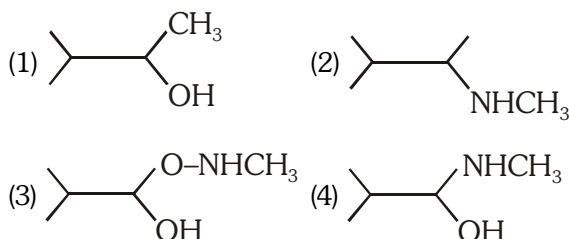
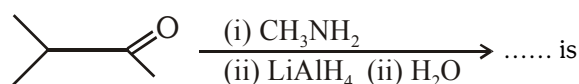
(1) a→r; b→t; c→p; d→q; e→s

(2) a→r; b→t; c→q; d→s; e→p

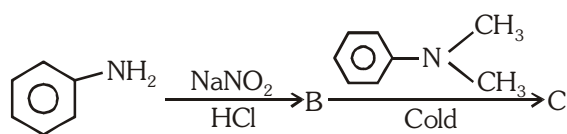
(3) a→q; b→s; c→p; d→r; e→t

(4) a→t; b→p; c→s; d→q; e→r

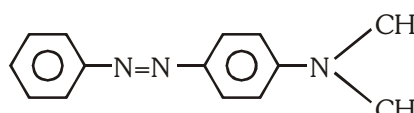
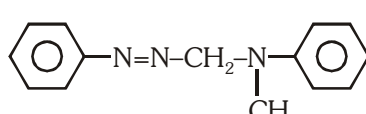
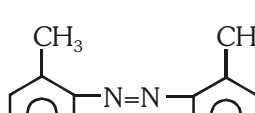
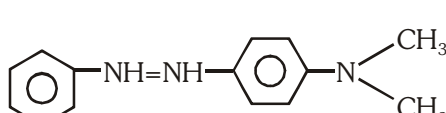
25. The major organic product formed from the following reaction :



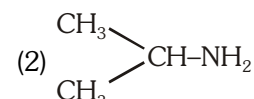
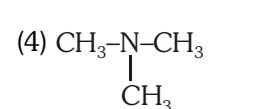
26. In a reaction of aniline a coloured product C was obtained.



The structure of C would be

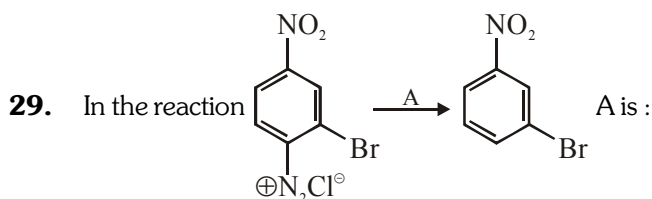
- (1) 
- (2) 
- (3) 
- (4) 

27. An organic compound ($\text{C}_3\text{H}_9\text{N}$) (A), when treated with nitrous acid gave an alcohol and N_2 gas was evolved. (A) on warming with CHCl_3 and caustic potash gave (C) which on reduction gave isopropylmethanamine. Predict the structure of (A) :

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ (2) 
- (3) $\text{CH}_3\text{CH}_2\text{NHCH}_3$ (4) 

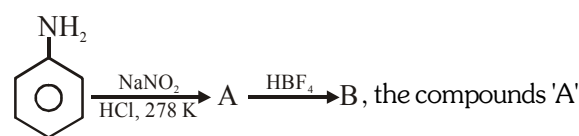
28. Method by which aniline cannot be prepared is :

- (1) Reduction of nitrobenzene with H_2/Pd in ethanol
- (2) Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with NaOH solution
- (3) Hydrolysis of phenylisocyanide with acidic solution
- (4) Degradation of benzamide with bromine in alkaline solution



- (1) $\text{H}^+/\text{H}_2\text{O}$ (2) $\text{HgSO}_4/\text{H}_2\text{SO}_4$
- (3) Cu_2Cl_2 (4) H_3PO_2 and H_2O

30. In the chemical reactions,



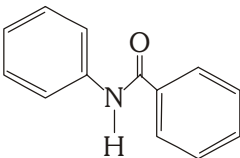
and 'B' respectively are :-

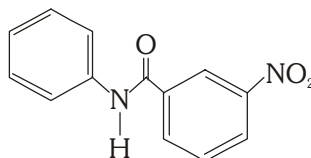
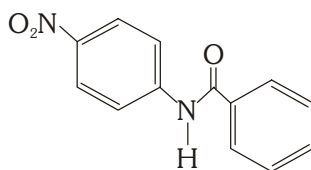
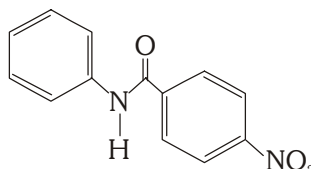
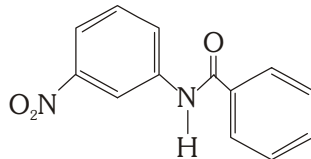
- (1) Nitrobenzene and chlorobenzene
- (2) Nitrobenzene and fluorobenzene
- (3) Phenol and benzene
- (4) Benzene diazonium chloride and fluorobenzene

				ANSWER KEY						Exercise-I	
Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	4	1	3	3	1	3	3	2	4	2	
Que.	11	12	13	14	15	16	17	18	19	20	
Ans.	2	3	1	2	2	3	1	2	2	2	
Que.	21	22	23	24	25	26	27	28	29	30	
Ans.	3	4	4	1	2	1	2	2	4	4	

13. Which one of the following is the strongest base in aqueous solution - [AIEEE-07]

- (1) Trimethylamine
- (2) Aniline
- (3) Dimethylamine
- (4) Methylamine

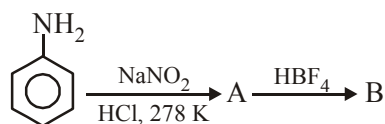
14. In the following reaction,  $\xrightarrow[\text{conc. H}_2\text{SO}_4]{\text{conc. HNO}_3}$ X, the structure of the major product 'X' is - [IIT-07]

- (1) 
- (2) 
- (3) 
- (4) 

15. Toluene is nitrated and the resulting product is reduced with tin and hydrochloric acid. The product so obtained is diazotized and then heated with cuprous bromide. The reaction mixture so formed contains [AIEEE-2008]

- (1) mixture of o- and p-bromotoluenes
- (2) mixture of o- and p-dibromobenzenes
- (3) mixture of o- and p-bromoanilines
- (4) mixture of o- and m-bromotoluenes

16. In the chemical reactions,

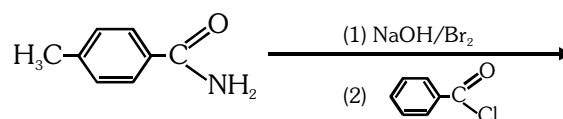


the compounds 'A' and 'B' respectively are :-

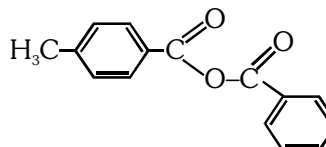
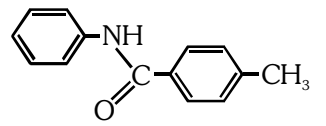
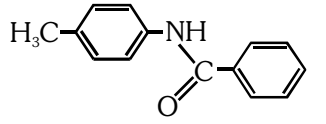
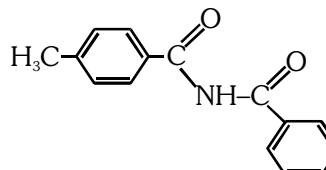
[AIEEE-2010]

- (1) Nitrobenzene and chlorobenzene
- (2) Nitrobenzene and fluorobenzene
- (3) Phenol and benzene
- (4) Benzene diazonium chloride and fluorobenzene

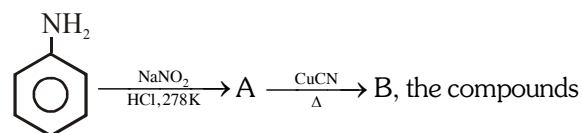
17. In the reaction



The structure of the product T is - [IIT-10]

- (1) 
- (2) 
- (3) 
- (4) 

18. In the chemical reactions

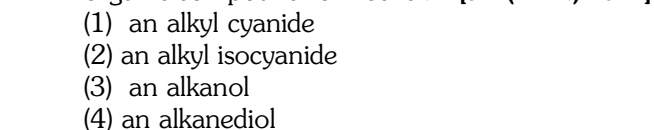


A and B respectively are :

[AIEEE-2011]

- (1) Fluorobenzene and phenol
- (2) Benzene diazonium chloride and benzonitrile
- (3) Nitrobenzene and chlorobenzene
- (4) Phenol and bromobenzene

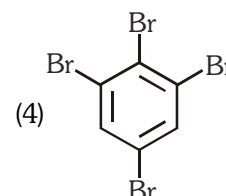
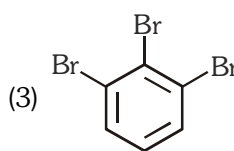
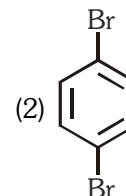
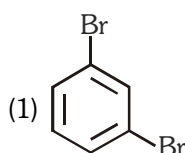
23. On heating an aliphatic primary amine with chloroform and ethanolic potassium hydroxide, the organic compound formed is :- **[JEE(MAIN)-2014]**



24. In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br₂ used per mole of amine produced are : **[JEE(MAIN)-2016]**

- (1) Four moles of NaOH and one mole of Br₂
- (2) One mole of NaOH and one mole of Br₂
- (3) Four moles of NaOH and two moles of Br₂
- (4) Two moles of NaOH and two moles of Br₂

25. The product(s) of the following reaction sequence is(are) **[JEE Adv. 2016]**



(1)  (2) 

(3)  (4) 

[JEE(MAIN)-2013]

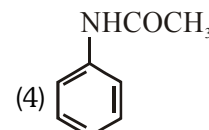
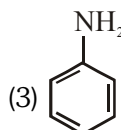
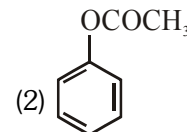
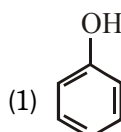
(1) 2 (2) 5 (3) 4 (4) 6

[JEE(MAIN)-2013]

(1) CH_3COOH (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (3) $\text{CH}_3\text{-}\underset{\text{CH}_3}{\text{CH}}\text{-COOH}$ (4) $\text{CH}_3\text{CH}_2\text{COOH}$

26. Which of the following compounds will significant amount of meta product during mono-nitration reaction ? **[JEE(MAIN)-2017]**

[JEE(MAIN)-2017]

56

BIOMOLECULES, POLYMER AND POC & CHEMISTRY IN EVERYDAY LIFE

EXERCISE-I

- 1.** Which one of the following metal ions is essential inside the cell for the metabolism of glucose/synthesis of proteins?

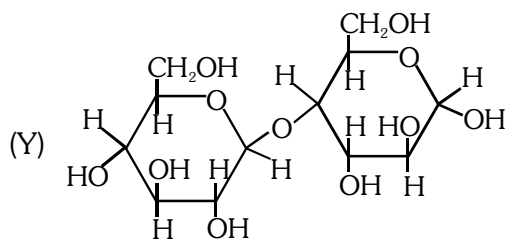
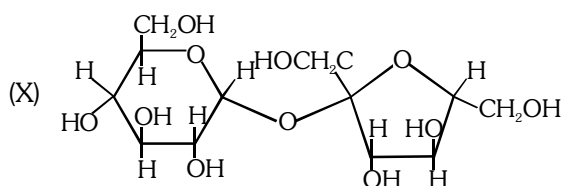
(1) Na^+

(2) K^+

(3) Mg^{2+}

(4) Ca^{2+}

- 2.** The correct statement about the following sugars X and Y is :

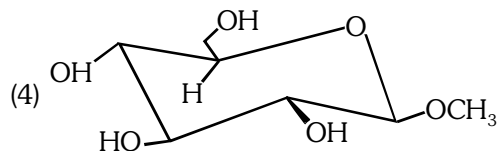
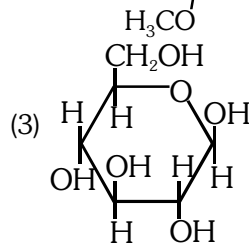
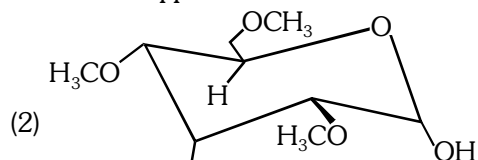
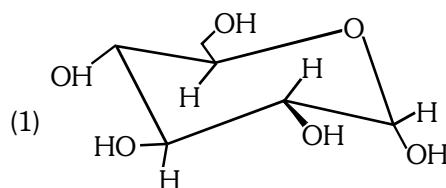


- (1) X is reducing sugar and Y is a non-reducing sugar
- (2) X is a non-reducing sugar and Y is reducing sugar
- (3) The glucosidic linkages in X and Y are α and α , respectively
- (4) The glucosidic linkage in X and Y are β and β , respectively

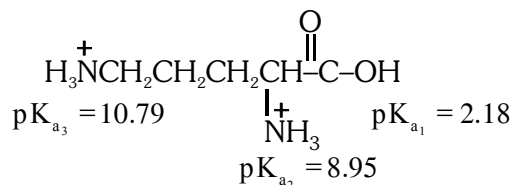
- 3.** Natural glucose is termed D-glucose because :

- (1) -OH on the X fifth carbon is on the right side in fischer projection
- (2) -OH on the second carbon is on the right side in fischer projection
- (3) -OH on the sixth carbon is on the right side in fischer projection
- (4) It is dextrorotatory

4. Identify the non reducing sugar



5. Lysine has $\text{pK}_{\text{a}_1} = 2.18$ and $\text{pK}_{\text{a}_2} = 8.95$. the pK_{a} for ionization of the side chain is 10.97



What is the isoelectric point of lysine ?

- (1) 5.56 (2) 6.48 (3) 9.87 (4) 10.83

6. The number of hydrogen bonds present in the sequence of a strand of a double helical DNA 5' ATGCCTAAG 3' is :

- (1) 24 (2) 19 (3) 22 (4) 20

- 7.** Mention true (T) and (F) out of the following.

S₁ : Sucrose gives negative tests with benedict's and Tollen's solutions.

S₂ : Sucrose does not form an osazone.

S₃ : Sucrose does not undergo mutarotation.

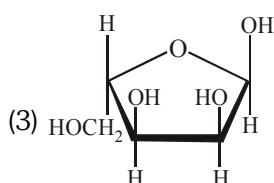
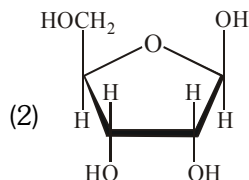
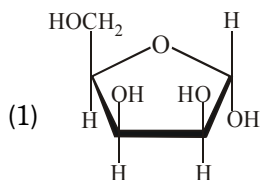
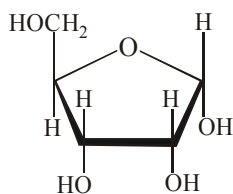
S₄ : Octamethyl derivative of sucrose, on hydrolysis, gives 2, 3, 4, 6-tetra-O-methyl-D-glucose and 1, 3, 4, 6-tetra-O-methyl D-fructose

S₅ : One mole of sucrose on acid hydrolysis yields one mole of D-glucose and one mole of D-fructose.

- (1) TTTTT (2) FTFTF (3) FFFTT (4) TFFFT

8. Which of the following carbohydrates are branched polymer of glucose :-
 (1) Amylose (2) Amylopectin
 (3) Cellulose (4) Glycogen
9. Select the correct statement among following :
 (1) Number of chiral atom in α -D-glucose is less than D-glucose
 (2) D-glucose and D-fructose give same product with HIO_4
 (3) D-glucose and D-fructose give same product with $\text{H}_2\text{N-OH}$
 (4) D-glucose and D-fructose form same product with $\text{H}_2\text{N-NH-Ph}$
10. Cellulose is a straight-chain polysaccharide composed of only :
 (1) D-Glucose units joined by α -glycosidic linkage
 (2) D-Glucose units joined by β -glycosidic linkage
 (3) D-Galactose units joined by α -glycosidic linkage
 (4) Galactose units joined by β -glycosidic linkage
11. The structures obtained on acidification (H^+) and basification (OH^-) of alanine yields respectively.
 (1) $\text{CH}_3-\underset{\text{NH}_3^+}{\text{CH}}-\text{COOH}$, $\text{CH}_3-\underset{\text{NH}_2}{\text{CH}}-\text{COO}^-$
 (2) $\text{CH}_3-\underset{\text{NH}_4^+}{\text{CH}}-\text{COO}^-$, $\text{CH}_3-\underset{\text{NH}_2\text{OH}}{\text{CH}}-\text{COO}^-$
 (3) $\text{CH}_3-\underset{\text{NH}_3^+}{\text{CH}_2}-\text{COO}^-$, $\text{CH}_3-\underset{\text{NH}}{\underset{\parallel}{\text{C}}}-\text{COO}^-$
 (4) $\text{CH}_3-\underset{\text{NH}_3^+}{\text{CH}_2}-\text{COO}^{2+}$, $\text{HOCH}_2-\underset{\text{NH}_3^+}{\text{CH}}-\text{COO}^-$
12. The vitamin which is water-soluble and is an antioxidant is :
 (1) vitamin E (2) vitamin B₁
 (3) vitamin C (4) vitamin D
13. The transfer RNA anticodon for the messenger RNA codon G-C-A is :
 (1) G-C-U (2) U-G-C (3) C-G-U (4) G-U-C
14. The metal ion which forms a violet-coloured complex with a protein in the presence of an alkali is :
 (1) Cu^{2+} (2) Zn^{3+} (3) Co^{3+} (4) Fe^{3+}
15. Which statement is incorrect about peptide bond ?
 (1) C-N bond length in proteins is longer than usual bond length of C-N bond
 (2) Spectroscopic analysis shows planar structure of of protein
 (3) C-N bond length in proteins is smaller than usual bond of C-N bond
 (4) None of these
16. Glucose on oxidation with nitric acid gives :
 (1) Gluconic acid (2) Saccharic acid
 (3) Sorbic acid (4) Aldonic acid
17. Which of the following sets contains only thermoplastics ?
 (1) Glyptal, Melmac, PAN
 (2) Polythene, Bakelite, Nylon-6
 (3) PVC, PMMA, Polystyrene
 (4) Polypropylene, Urea-formaldehyde, Teflon
18. Given the polymers, A = Nylon 6-6; B = Buna-S; C = Polythene. Arrange these in decreasing order of their intermolecular forces.
 (1) $A > C > B$ (2) $B > C > A$
 (3) $A > B > C$ (4) $C > B > A$
19. Which of the following monomers can undergoes radical, cationic as well as anionic polymerization with equal ease ?
 (1) $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{C}}}=\text{CH}_2$ (2) $\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$
 (3) $\text{CH}_2=\text{CH}-\text{CN}$ (4) $\text{CH}_2=\text{CH}_2$
20. The chemical name for melamine is -
 (1) 1, 3, 5-Triamino-2, 4, 6-triazine
 (2) 2, 4, 6-Triamino-1, 3, 5-triazine
 (3) 2-Amino-1, 3, 5-triazine
 (4) 3, 4-Diamino-1, 3, 5-triazine
21. Which of the following is not correctly matched ?
 (1) Nylon-6,6: $\left[\text{NH}-(\text{CH}_2)_6-\text{NH}-\text{CO}-(\text{CH}_2)_4-\overset{\text{O}}{\parallel}{\text{C}} \right]_n$
 (2) Neoprene: $\left[\text{CH}_2-\underset{\text{Cl}}{\underset{|}{\text{C}}}=\text{CH}-\text{CH}_2 \right]_n$
 (3) Terylene: $\left[\text{OCH}_2-\text{CH}_2-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_4-\overset{\text{O}}{\parallel}{\text{C}} \right]_n$
 (4) PMMA: $\left[\text{CH}_2-\underset{\text{COOCH}_3}{\underset{|}{\text{C}}}(\text{CH}_3) \right]_n$

22. Which of the following represents the anomer of the compound shown ?



(4) None of these

23. Biodegradable polymer which can be produced from glycine and aminocaproic acid is

- (1) PHBC (2) Buna-N
(3) Nylon-6, 6 (4) Nylon 2-nylon 6

24. Which of the following chemicals can be added for sweetening of food items at cooking temperature and does not provide calories ?

- (1) Sucrose (2) Glucose
(3) Aspartame (4) Sucralose

25. Which of the following is an example of liquid dishwashing detergent ?

- (1) $\text{CH}_3(\text{CH}_2)_{10}-\text{CH}_2\text{OSO}_3^-\text{Na}^+$
(2) $\text{C}_9\text{H}_{19}-\text{C}_6\text{H}_4-\text{O}-(\text{CH}_2-\text{CH}_2-\text{O})_5-\text{CH}_2-\text{CH}_2\text{OH}$
(3) $\text{CH}_3-\text{C}_6\text{H}_4-\text{SO}_3^-\text{Na}^+$

- (4) $\left[\text{CH}_3-(\text{CH}_2)_{15}-\text{N}(\text{CH}_3)_3 \right]^+ \text{Br}^-$

26. Dettol, an antiseptic, consists of :

- (1) cresol and ethanol
(2) xynol and
(3) chloroxynol and terpineol
(4) phenol and cresol

27. Select the correct statements (s).

- (1) The antibiotics which are effective mainly against gram-positive or gram-negative bacteria are called narrow-spectrum antibiotics
(2) The antibiotics which kill or inhibit a wide range of gram-positive or gram-negative bacteria are called broad-spectrum antibiotics
(3) Penicillin G is a narrow-spectrum antibiotics while ofloxacin is a broad-spectrum antibiotics
(4) All of the above

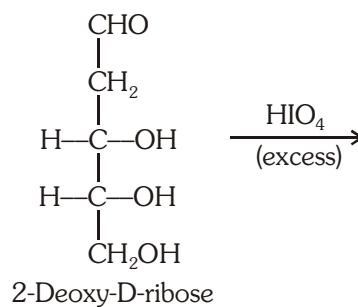
28. Starch molecules are polymer with repeating glucose units. Select the incorrect statement.

- (1) Glucose units are joined through α -glycosidic linkage
(2) The branches of amylopectin are linked to the chain with α -1,6'-glycosidic linkages
(3) The linear linkages of amylopectin are formed by α -1,6'-glycosidic bond
(4) Amylose has an unbranched skeleton of glucose molecules with α -1,4'-glycosidic linkages

29. Which is not the correct matching of medicine with its disease/activity ?

- (1) Aspirin-pain reliever
(2) Equanil-hypertension
(3) Chloramphenicol-typhoid
(4) 0.2 per cent phenol-disinfectant

30. The products expected from the reaction are :



- (1) OHCCH_2CHO , HCOOH and HCHO
(2) HOOCCHOHCHOHCOOH and HCOOH
(3) $\text{HOOCCH}_2\text{COOH}$, HCOOH and HCHO
(4) $\text{HOOCCH}_2\text{COOH}$, HCOOH and HCHO

ANSWER KEY

Exercise-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	4	4	3	3	1	2	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	3	1	3	2	3	1	2	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	4	4	2	3	4	3	4	1

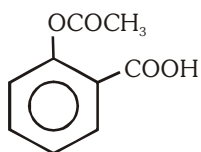
PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. A substance forms Zwitter ion. It has functional groups- [AIEEE - 2002]

- (1) $-\text{NH}_2$, $-\text{COOH}$
 (2) $-\text{NH}_2$, $-\text{SO}_3\text{H}$
 (3) Both (1) and (2)
 (4) None of these

2. Compound A given below is :



[AIEEE - 2002]

- (1) Antiseptic (2) Antibiotic
 (3) Analgesic (4) Pesticide

3. RNA contains- [AIEEE - 2002]

- (1) Ribose sugar and thymine
 (2) Ribose sugar and uracil
 (3) deoxyribose sugar and uracil
 (4) deoxyribose sugar and thymine

4. Monomers are converted to polymer by- [AIEEE - 2002]

- (1) Hydrolysis of monomer
 (2) Condensation reaction between monomers
 (3) protonation of monomers
 (4) none of the above

5. Complete hydrolysis of cellulose gives- [AIEEE - 2003]

- (1) D-fructose (2) D-ribose
 (3) D-glucose (4) L-glucose

6. The reason for double helical structure of DNA is an operation of- [AIEEE - 2003]

- (1) Vanderwaal's forces
 (2) Dipole-Dipole interaction
 (3) Hydrogen bonding
 (4) Electrostatic attractions

7. Nylon threads are made of- [AIEEE - 2003]

- (1) polyvinyl polymer
 (2) polyester polymer
 (3) polyamide
 (4) polyethylene polymer

8. Which base is present in RNA but not in DNA [AIEEE - 2004]

- (1) Uracil (2) Cytosine
 (3) Guanine (4) Thymine

9. Insulin production and its action in human body are responsible for the level of diabetes. This compound belongs to which of the following categories [AIEEE - 2004]

- (1) A coenzyme (2) A hormone
 (3) An enzyme (4) An antibiotic

10. Which one of the following types of drug reduces fever- [AIEEE - 2005]

- (1) Tranquilizer (2) Antibiotic
 (3) Antipyretic (4) Analgesic

11. Which of the following is a polyamide [AIEEE - 2005]

- (1) Bakelite (2) Terylene
 (3) Nylon-66 (4) Teflon

12. In both DNA and RNA, heterocyclic base and phosphate ester linkage are at- [AIEEE - 2005]

- (1) C_5' and C_1' respectively of the sugar molecule
 (2) C_1' and C_5' respectively of the sugar molecule
 (3) C_2' and C_5' respectively of the sugar molecule
 (4) C_5' and C_2' respectively of the sugar molecule

13. Which of the following is fully fluorinated polymer- [AIEEE - 2005]

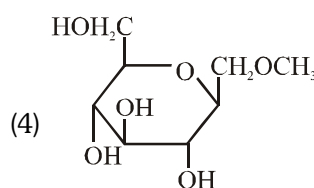
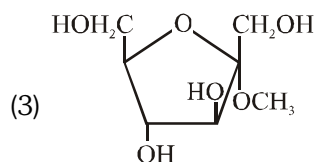
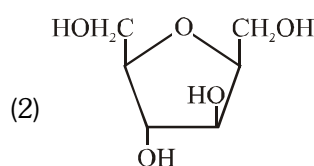
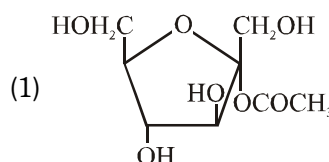
- (1) PVC (2) Thiokol
 (3) Teflon (4) Neoprene

14. The term anomers of glucose refers to- [AIEEE - 2006]

- (1) Isomers of glucose that differ in configurations at carbon one and four (C-1 and C-4)
 (2) A mixture of (D)-glucose and (L)-glucose
 (3) Enantiomers of glucose
 (4) Isomers of glucose that differ in configuration at carbon one (C-1)

- 15.** The pyrimidine bases present in DNA are- [AIEEE - 2006]
 (1) cytosine and adanine
 (2) cytosine and guanine
 (3) cytosine and thymine
 (4) cytosine and uracil
- 16.** The secondary structure of a protein refers to- [AIEEE - 2007]
 (1) α -helical backbone
 (2) hydrophobic interactions
 (3) sequence of α -amino acids
 (4) fixed configuration of the polypeptide backbone
- 17.** Buna-N synthetic rubber is a copolymer of :- [AIEEE-2009]
 (1) $\text{H}_2\text{C} = \text{CH} - \text{CN}$ and $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$
 (2) $\text{H}_2\text{C} = \text{CH} - \text{CN}$ and $\text{H}_2\text{C} = \text{CH} - \underset{\text{CH}_3}{\text{C}} = \text{CH}_2$
 (3) $\text{H}_2\text{C} = \text{CH} - \overset{\text{Cl}}{\text{C}} = \text{CH}_2$ and $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$
 (4) $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$ and $\text{H}_5\text{C}_6 - \text{CH} = \text{CH}_2$
- 18.** The two functional groups present in a typical carbohydrate are :- [AIEEE-2009]
 (1) $>\text{C} = \text{O}$ and $-\text{OH}$
 (2) $-\text{OH}$ and $-\text{CHO}$
 (3) $-\text{OH}$ and $-\text{COOH}$
 (4) $-\text{CHO}$ and $-\text{COOH}$
- 19.** The polymer containing strong intermolecular forces e.g. hydrogen bonding, is :- [AIEEE-2010]
 (1) natural rubber (2) teflon
 (3) nylon 6, 6 (4) polystyrene
- 20.** Biuret test is not given by :- [AIEEE-2010]
 (1) proteins (2) carbohydrates
 (3) polypeptides (4) urea
- 21.** The presence or absence of hydroxy group on which carbon atom of sugar differentiates RNA and DNA? [AIEEE-2011]
 (1) 3rd (2) 4th
 (3) 1st (4) 2nd
- 22.** Which one of the following bases is **not present** in DNA ? [JEE(Main)-2014]
 (1) Cytosine (2) Thymine
 (3) Quinoline (4) Adenine
- 23.** Which one is classified as a condensation polymer? [JEE(Main)-2014]
 (1) Teflon (2) Acrylonitrile
 (3) Dacron (4) Neoprene

- 24.** Thiol group is present in :- [JEE (Main)-2016]
 (1) Methionine (2) Cytosine
 (3) Cystine (4) Cysteine
- 25.** Which of the following statements about low density polythene is FALSE ? [JEE (Main)-2016]
 (1) It is used in the manufacture of buckets, dust-bins etc.
 (2) Its synthesis requires high pressure
 (3) It is a poor conductor of electricity
 (4) Its synthesis requires dioxygen or a peroxide initiator as a catalyst.
- 26.** Which of the following is an anionic detergent ? [JEE (Main)-2016]
 (1) Glyceryl oleate
 (2) Sodium stearate
 (3) Sodium lauryl sulphate
 (4) Cetyltrimethyl ammonium bromide
- 27.** The formation of which of the following polymers involves hydrolysis reaction ? [JEE (Main)-2017]
 (1) Nylon 6 (2) Bakelite
 (3) Nylon 6, 6 (4) Terylene
- 28.** Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution ? [JEE (Main)-2017]



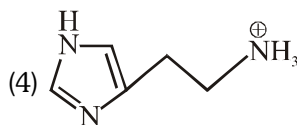
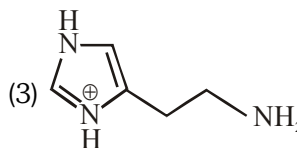
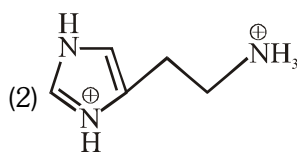
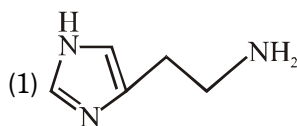
29. The formation of which of the following polymers involved hydrolysis reaction ?

[JEE (Main)-2017]

- (1) Nylon 6
- (2) Bakelite
- (3) Nylon 6, 6
- (4) Terylene

30. The predominant form of histamine present in human blood is (pK_a , Histidine = 6.0)

[JEE (Main)-2018]



PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	2	2	3	3	3	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	3	4	3	1	1	1	3	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	3	3	4	1	3	1	1	1	1